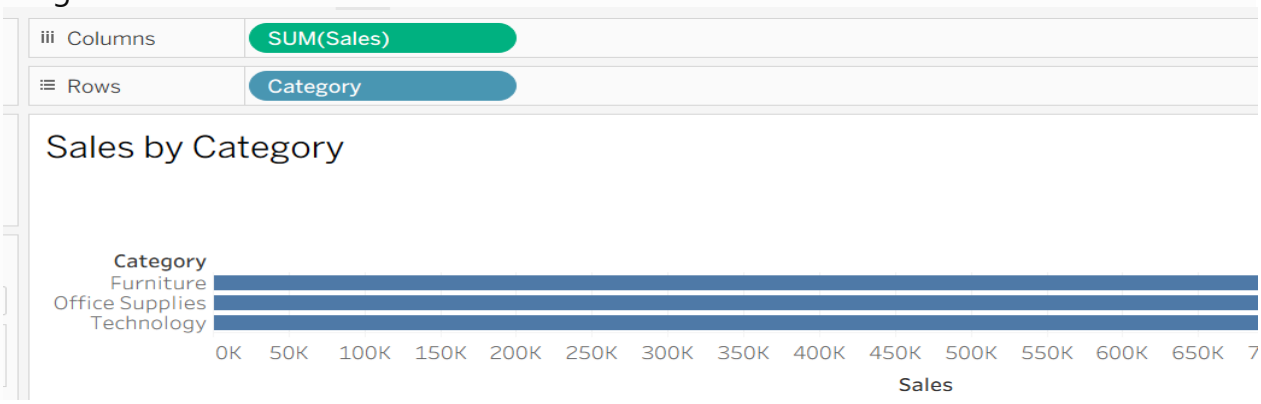
2(a) Design a Basic Dashboard Layout Using Sheets in Tableau

Aim: To combine multiple worksheets into a single dashboard in Tableau and arrange them in a clear, readable layout.

Steps to perform

1. Open Tableau Desktop and connect to the **Sample Superstore** dataset.
2. Create a worksheet named **Sales by Category**.

- Drag **Category** to Rows.
- Drag **Sales** to Columns.



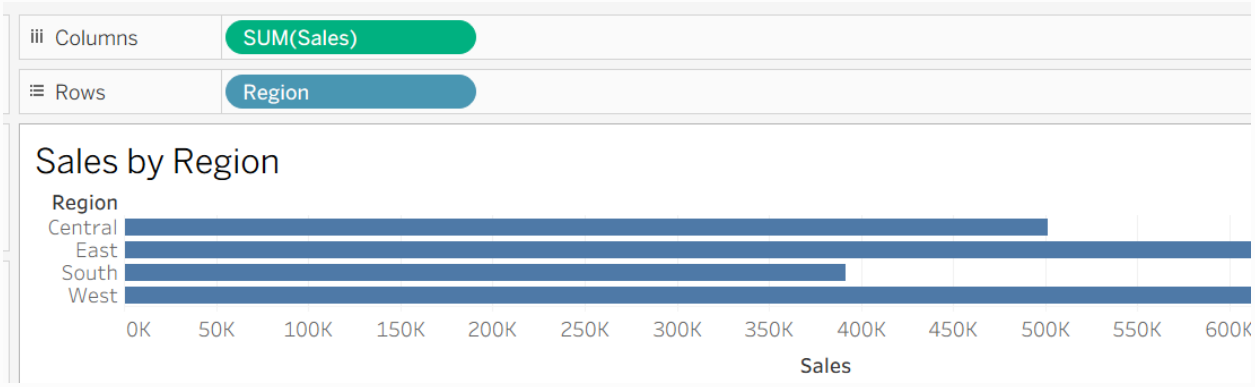
1. Create a worksheet named **Monthly Sales Trend**.

- Drag **Order Date** to Columns.
- Select **Month (Continuous)**.
- Drag **Sales** to Rows.
- Select **Line Chart** from Show Me.



Create a worksheet named **Sales by Region**.

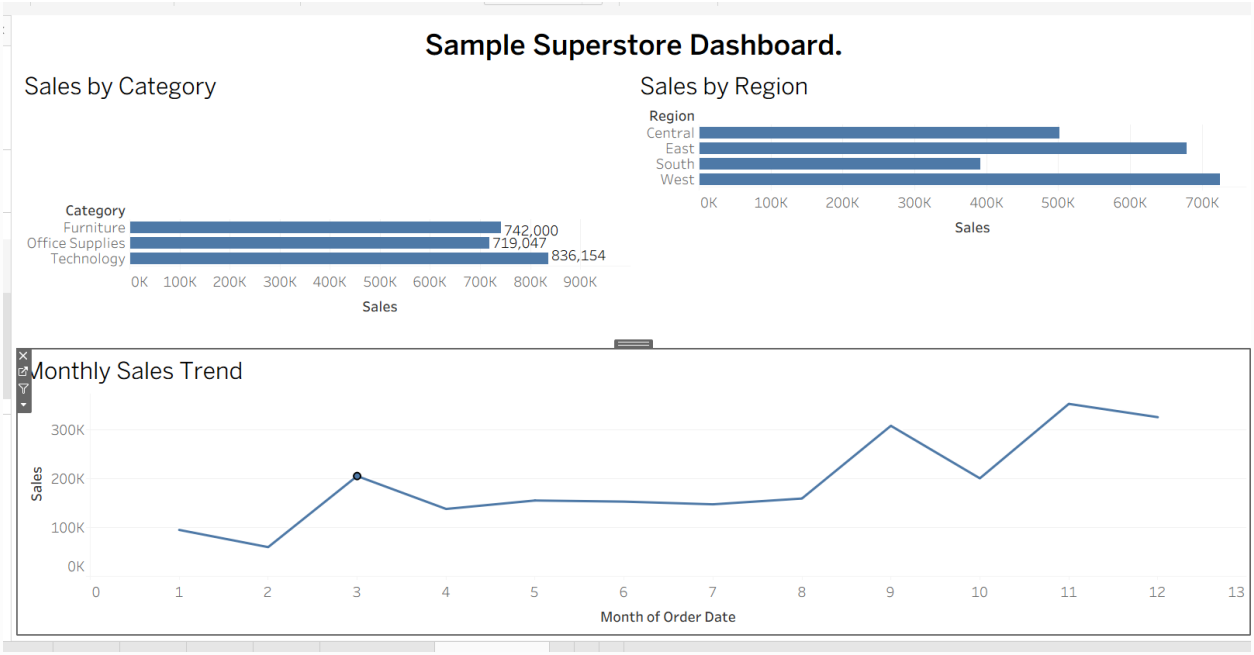
- Drag **Region** to Rows.
- Drag **Sales** to Columns.
- Choose a bar chart.



1. Click the **New Dashboard** icon at the bottom of Tableau.
2. In the Dashboard pane, choose a suitable size, such as **Automatic**.
3. Drag the three worksheets onto the dashboard.
4. Arrange the worksheets neatly:

- Top-left: Sales by Category.
- Top-right: Monthly Sales Trend.
- Bottom: Sales by Region.

5. Add a suitable dashboard title, such as *Sample Superstore Dashboard*



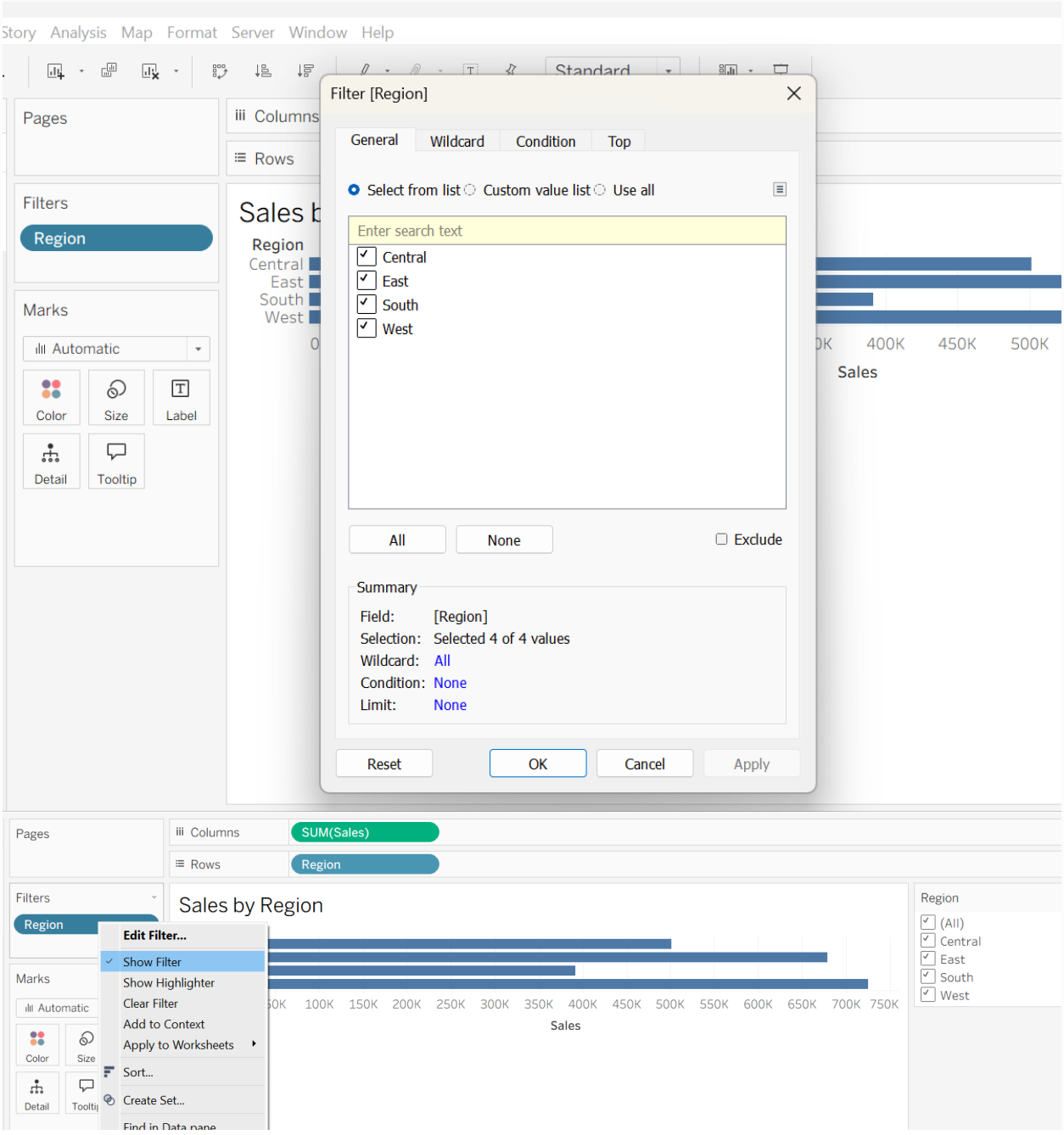
2(b) Add Filters and Interactive Controls in Tableau

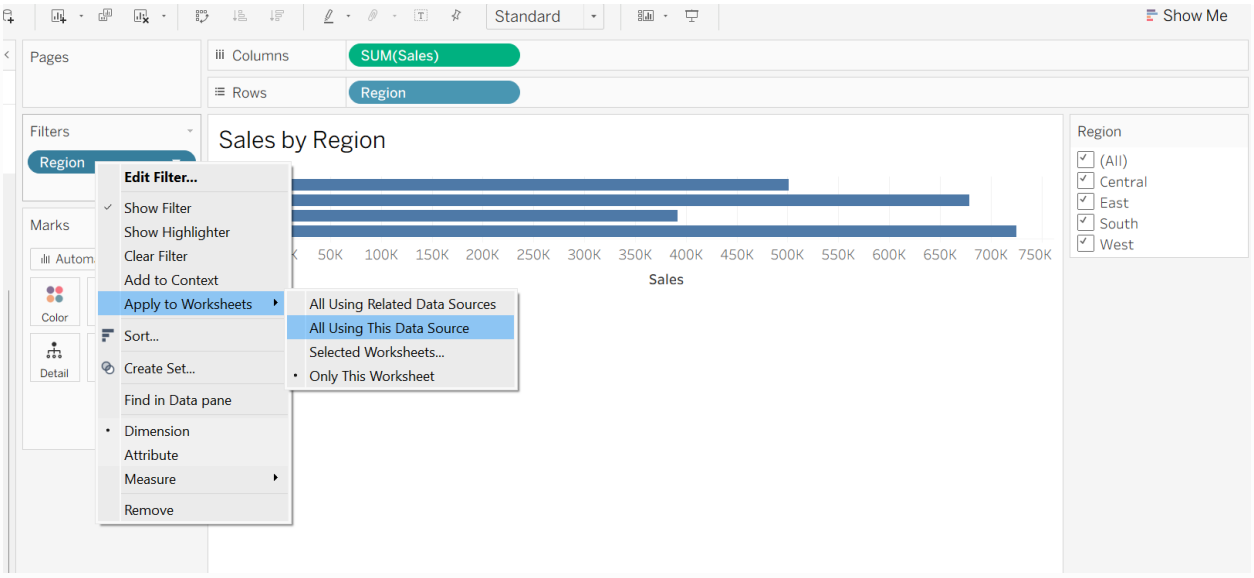
Aim: To add filters and interactive controls to a dashboard so that users can explore data dynamically.

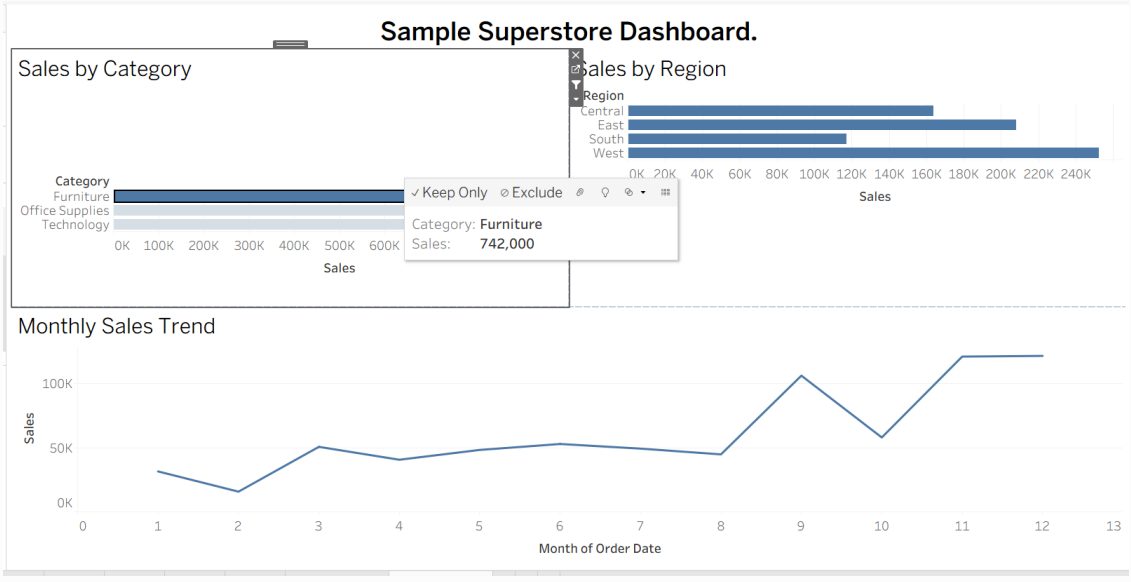
Steps to perform

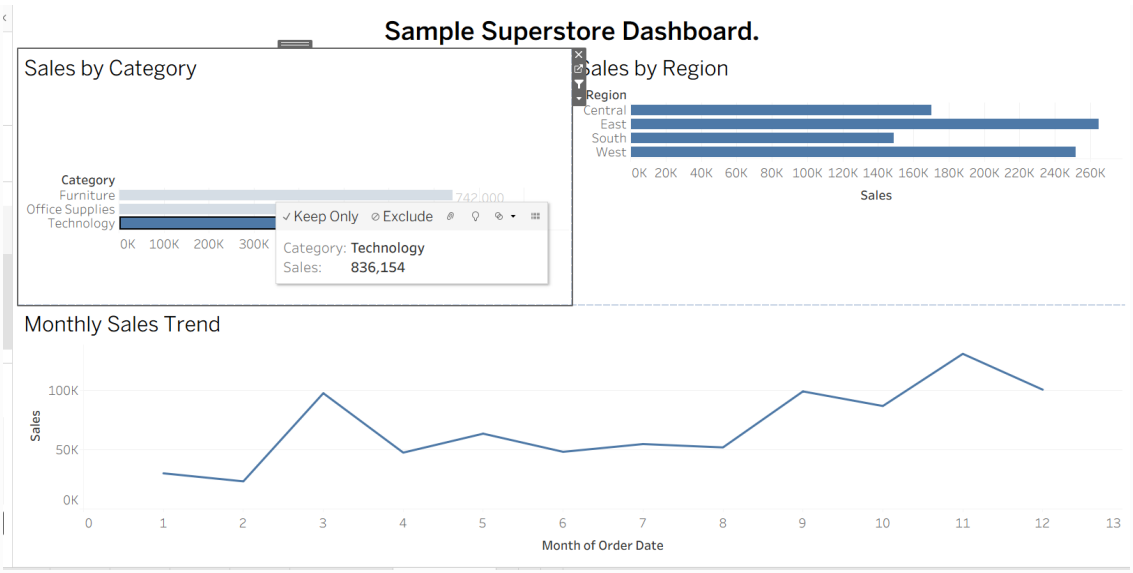
1. Open the **Sales by Region** worksheet or another worksheet where you want to add a filter.
2. Drag the **Region** field to the Filters shelf.
3. Select the required regions and click **OK**.
4. Right-click the Region field on the Filters shelf and select **Show Filter**. The Region filter card will appear.
5. Return to the dashboard by clicking its tab.
6. If the Region filter is not visible, open the worksheet's drop-down menu on the dashboard and select **Filters → Region**.
7. On the Region filter card's drop-down menu, choose **Apply to Worksheets → All Using This Data Source**. This makes the filter affect the connected worksheets using the same data source.
8. To make a chart interactive, select the chart on the dashboard and click **Use as Filter** (the funnel icon).

9. Test the dashboard by selecting a region in the filter or clicking a bar in a chart. The connected views should update based on the selection.
10. Format the filter card as a drop-down list or single-value selector if desired, and adjust the dashboard layout.




Standard
Show Me
Pages
Columns
SUM(Sales)
Rows
Region
Filters
Region
Sales by Region
Edit Filter...
Show Filter
Show Highlighter
Clear Filter
Add to Context
Apply to Worksheets
Sort...
Create Set...
Find in Data pane
Dimension
Attribute
Measure
Remove
All Using Related Data Sources
All Using This Data Source
Selected Worksheets...
Only This Worksheet
Marks
Color
Detail
50K 100K 150K 200K 250K 300K 350K 400K 450K 500K 550K 600K 650K 700K 750K
Sales
Region
(All)
Central
East
South
West


Sample Superstore Dashboard.
Sales by Category
Category
Furniture
Office Supplies
Technology
0K 100K 200K 300K 400K 500K 600K
Sales
Keep Only
Exclude
Category: Furniture
Sales: 742,000
Sales by Region
Region
Central
East
South
West
0K 20K 40K 60K 80K 100K 120K 140K 160K 180K 200K 220K 240K
Sales
Monthly Sales Trend
100K
50K
0K
Sales
0 1 2 3 4 5 6 7 8 9 10 11 12 13
Month of Order Date


Sample Superstore Dashboard.
Sales by Category
Category
Furniture
Office Supplies
Technology
742,000
0K 100K 200K 300K
Keep Only
Exclude
Category: Technology
Sales: 836,154
Sales by Region
Region
Central
East
South
West
0K 20K 40K 60K 80K 100K 120K 140K 160K 180K 200K 220K 240K 260K
Sales
Monthly Sales Trend
Sales
100K
50K
0K
0 1 2 3 4 5 6 7 8 9 10 11 12 13
Month of Order Date

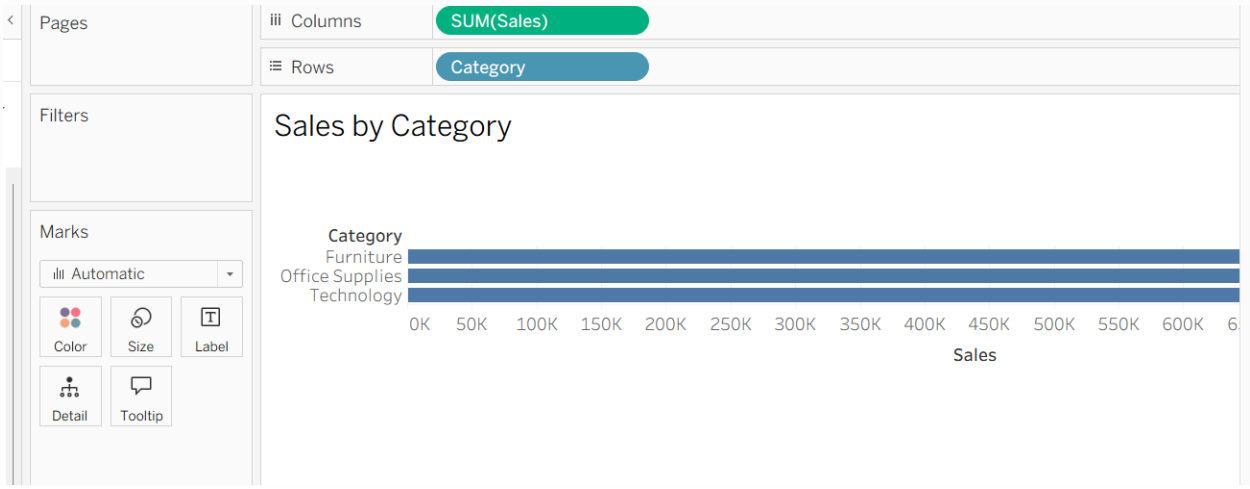
Practical 6: Formatting in Tableau

Aim: To apply worksheet-level and workbook-level formatting and customize number, date, and null value formats.

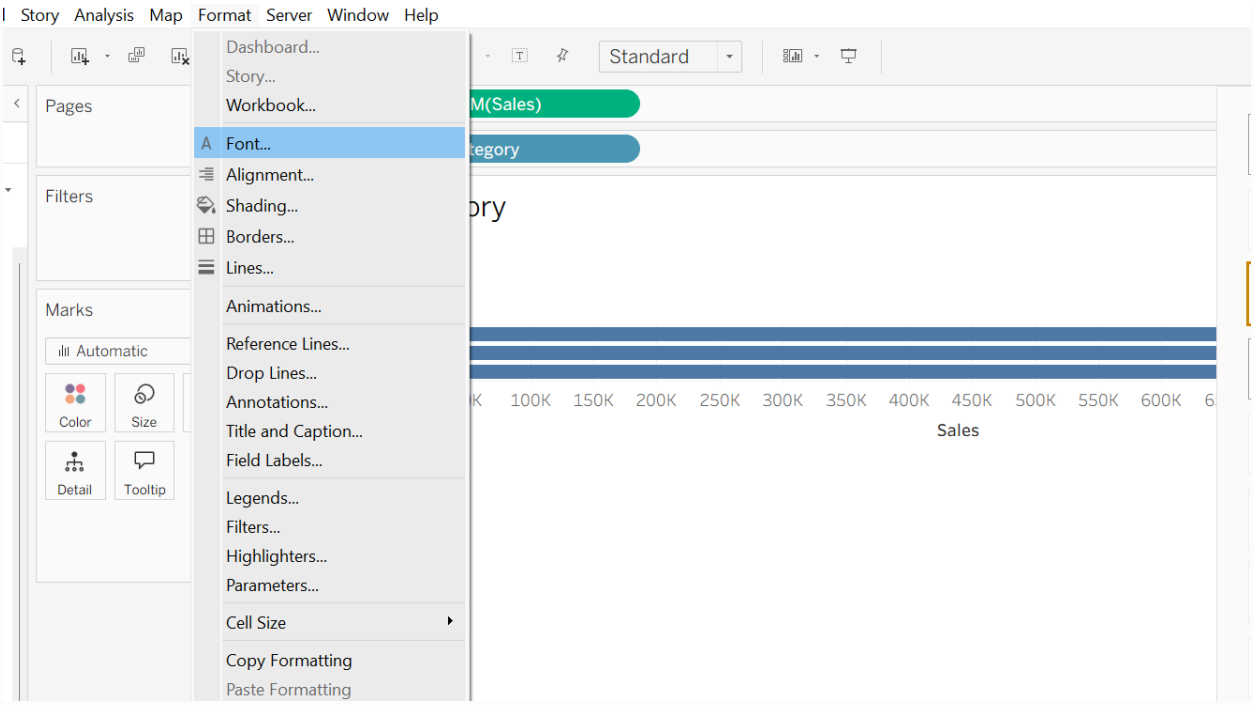
6(a) Apply Worksheet-Level and Workbook-Level Formatting

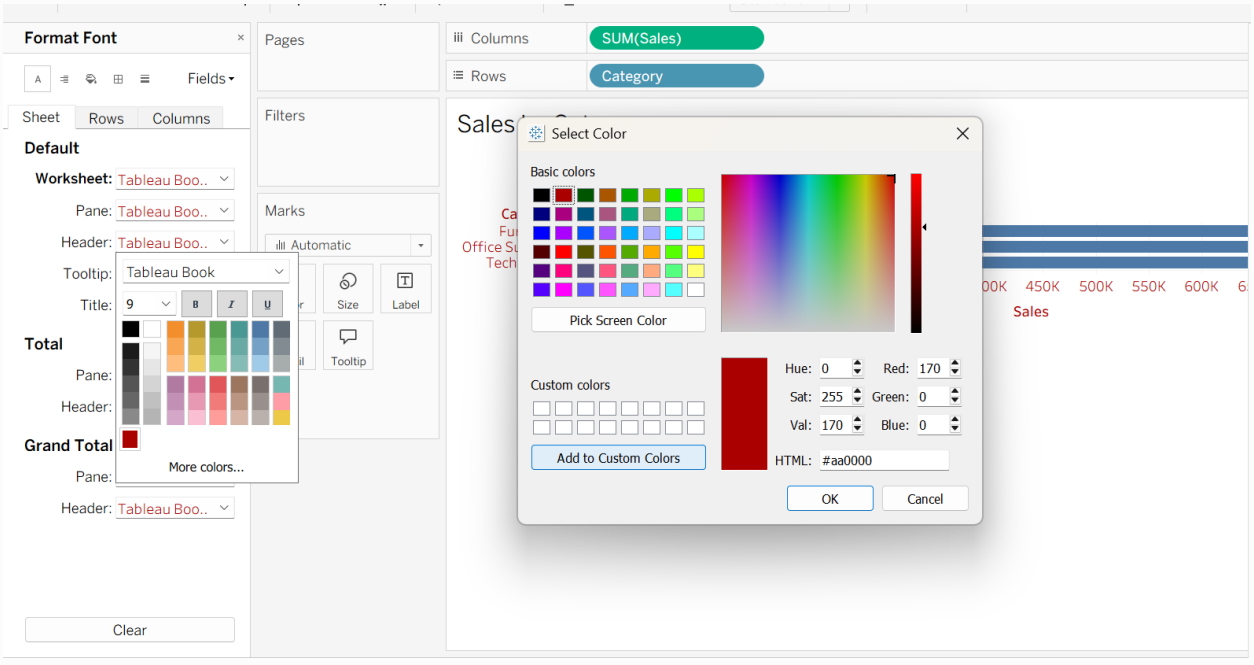
1.
2. Open your existing Tableau workbook containing the Sample Superstore dashboard.
3. Select a worksheet, such as **Sales by Category**.

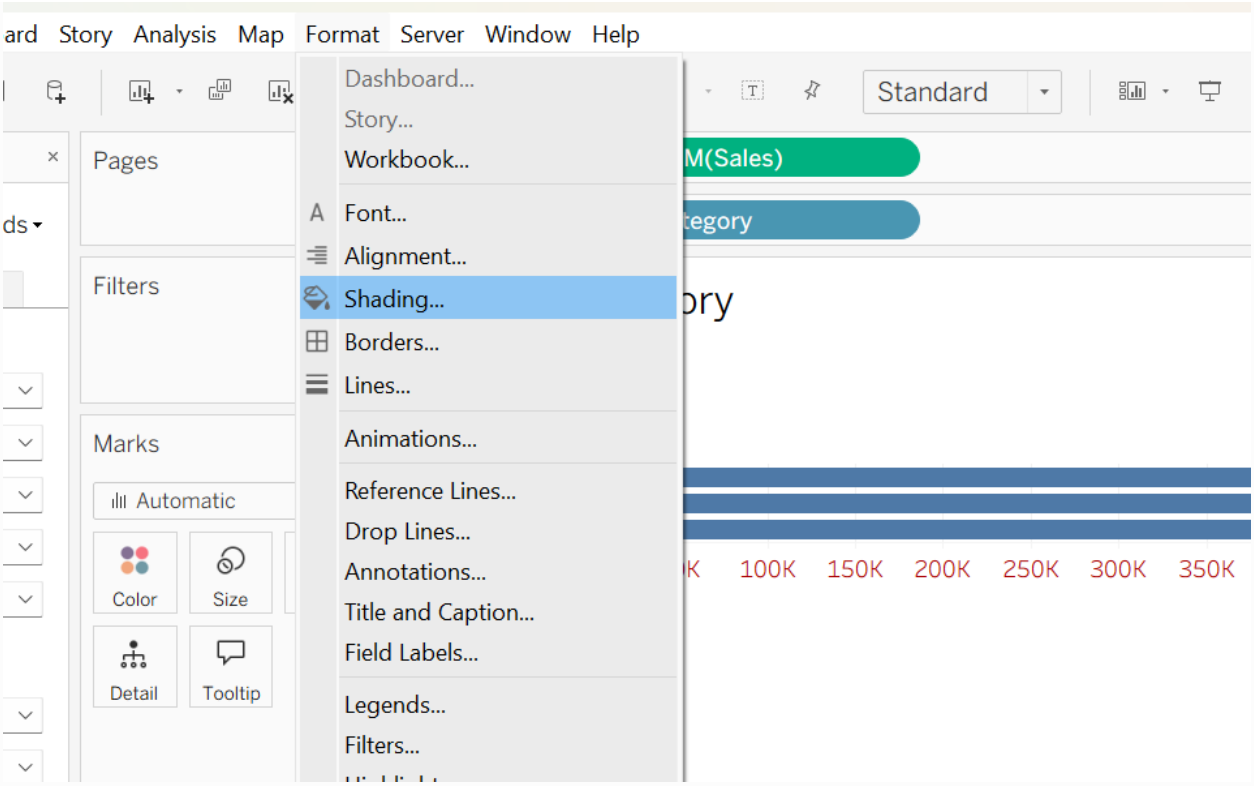
Sales by Category: drag Category to Rows and Sales to Columns, select Bar Chart and rename the sheet.

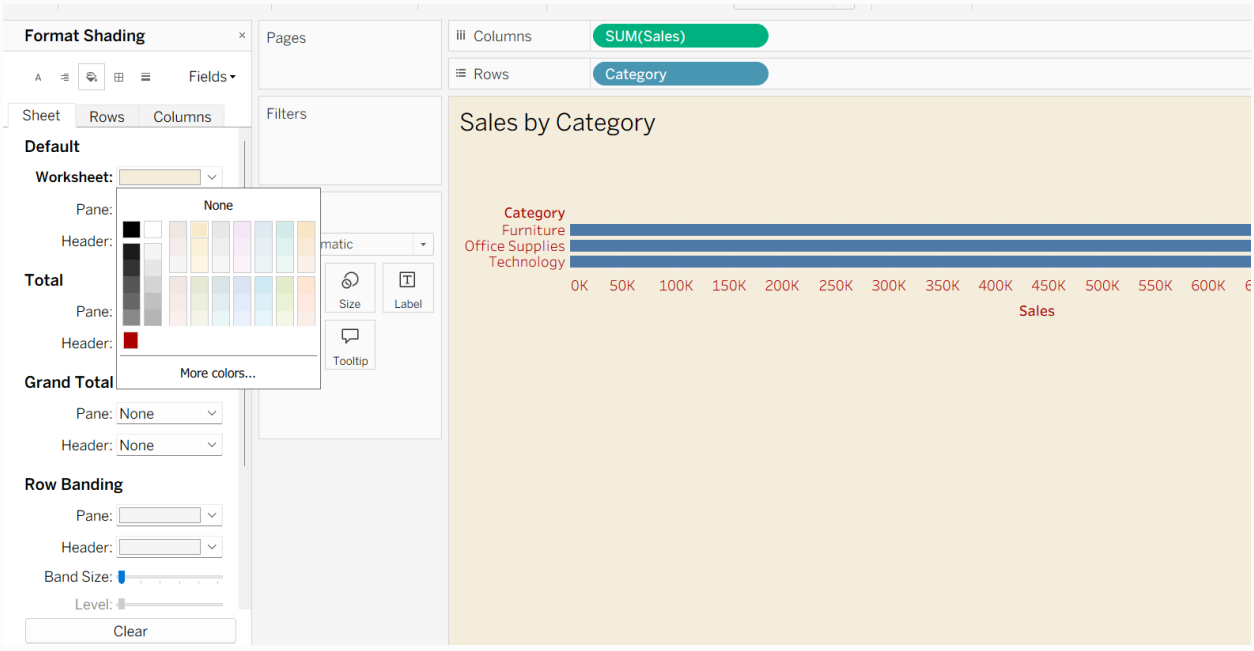


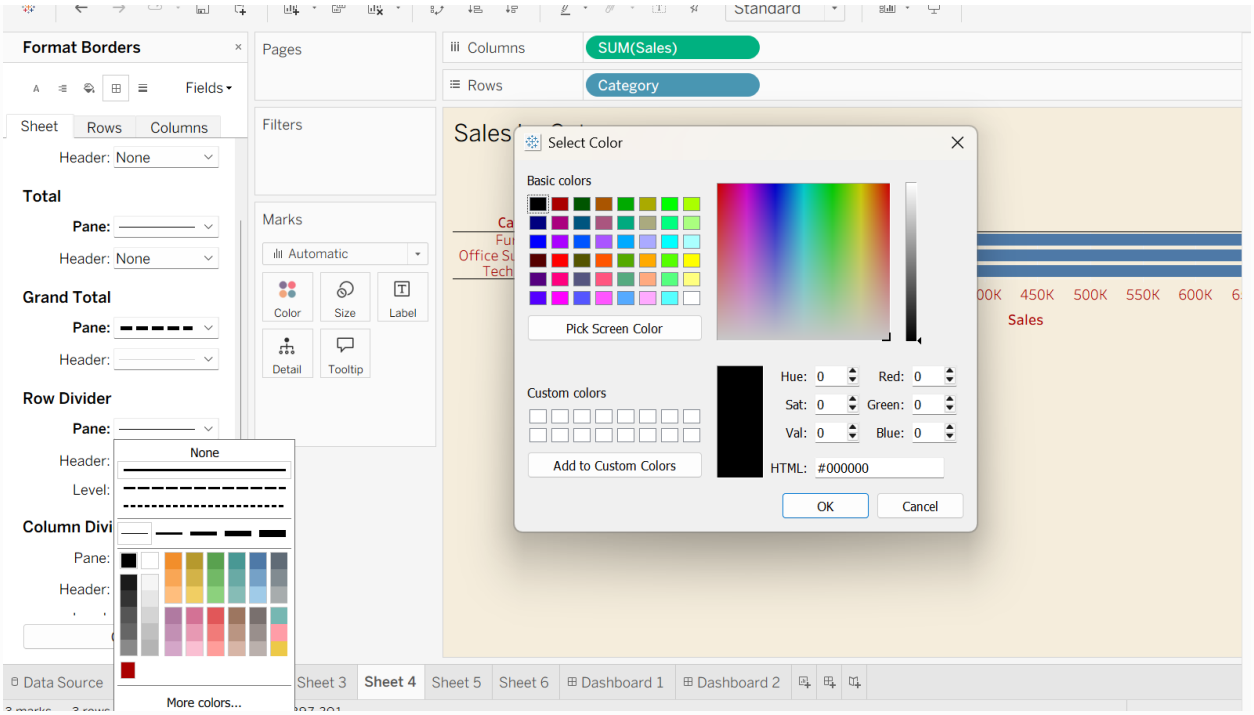
1. **Format the title:** Double-click the worksheet title and set it to `Sales by Category`. Make it bold and increase the font size.
2. **Format the font:** Go to **Format** → **Font** and adjust the worksheet font and header font.
3. **Format the background:** Go to **Format** → **Shading** and choose a suitable background color.
4. **Format the borders:** Go to **Format** → **Borders** and adjust the dividers if needed.

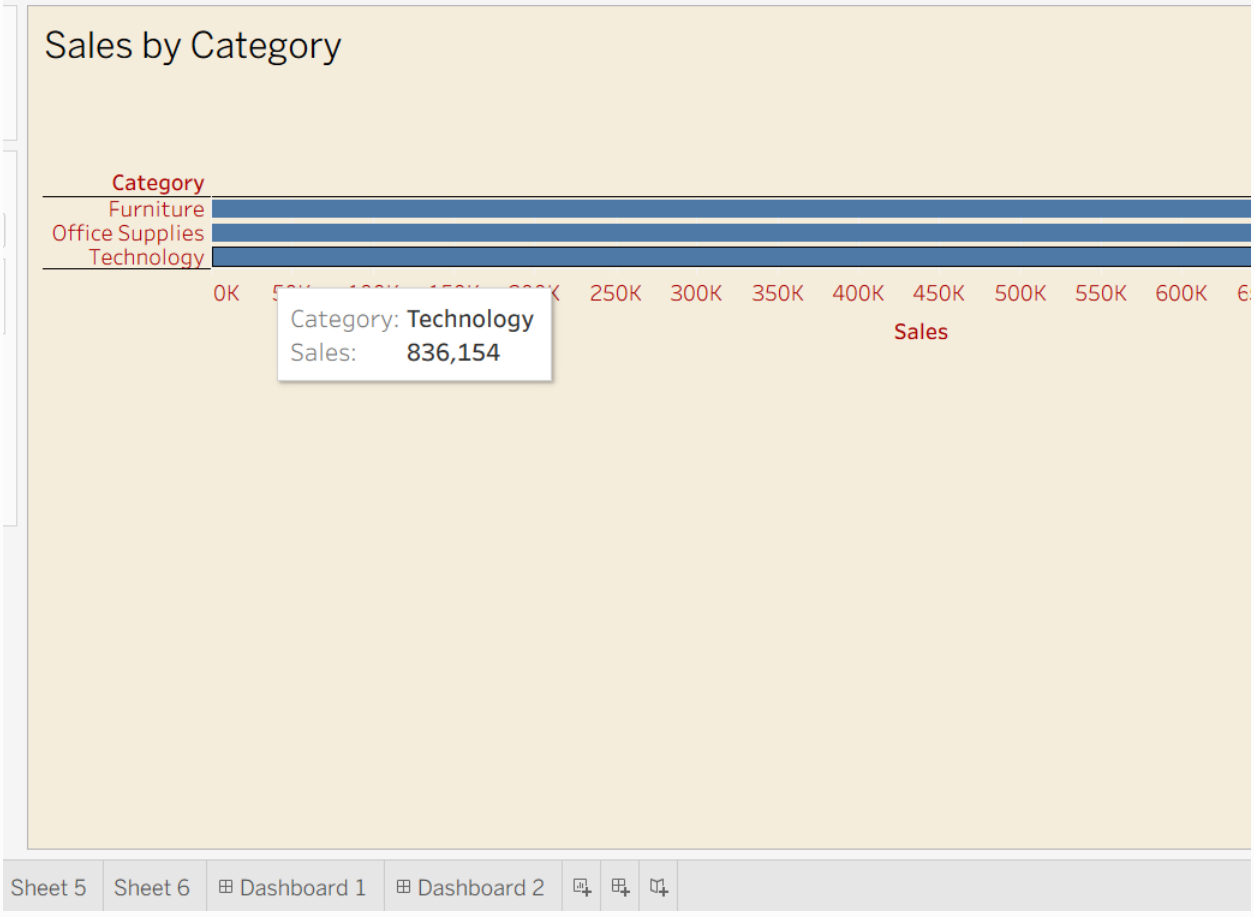

Story Analysis Map Format Server Window Help
Dashboard...
Story...
Workbook...
Font...
Alignment...
Shading...
Borders...
Lines...
Animations...
Reference Lines...
Drop Lines...
Annotations...
Title and Caption...
Field Labels...
Legends...
Filters...
Highlighters...
Parameters...
Cell Size
Copy Formatting
Paste Formatting
Standard
Pages
Filters
Marks
Automatic
Color
Size
Detail
Tooltip
100K 150K 200K 250K 300K 350K 400K 450K 500K 550K 600K
Sales

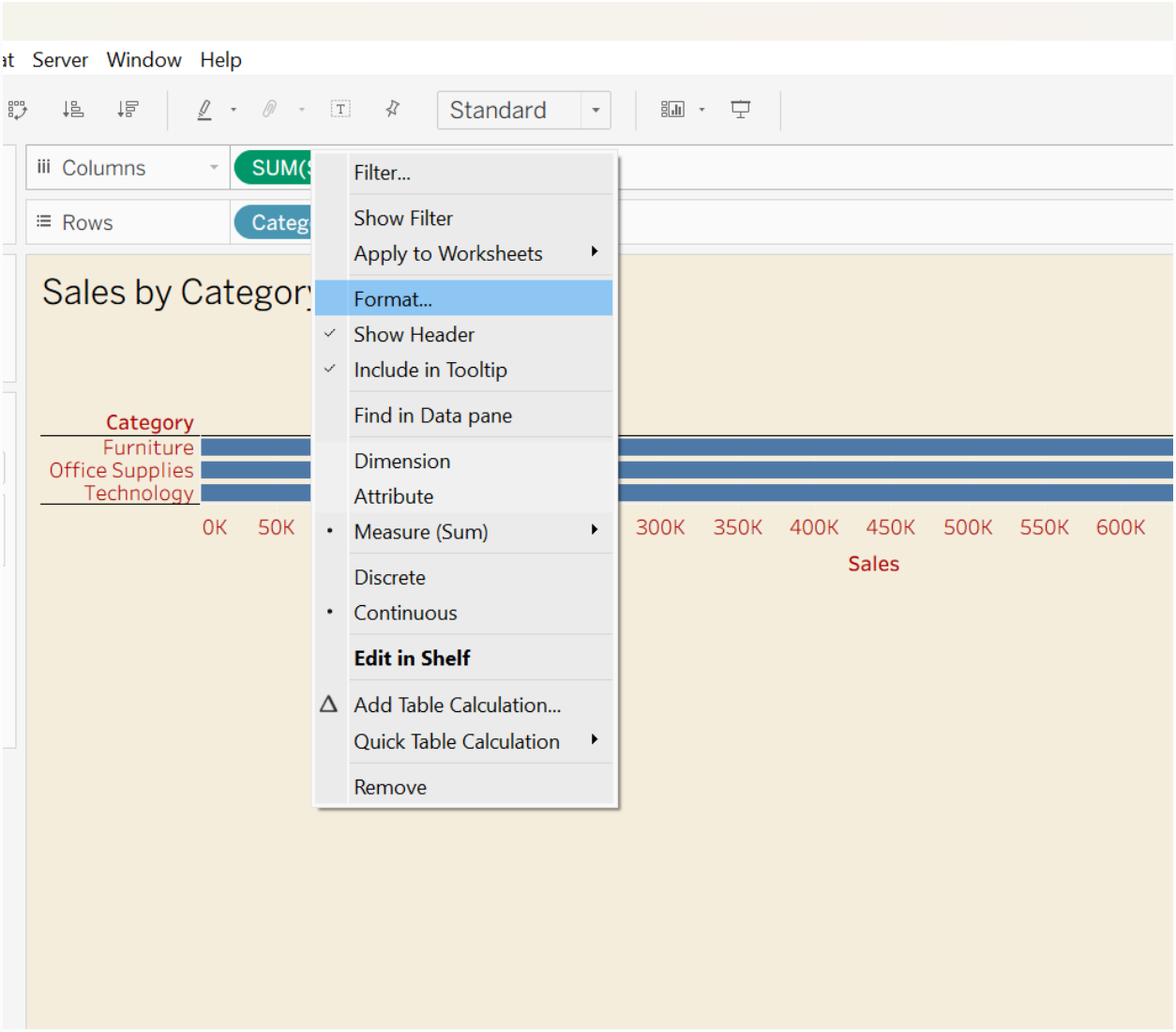

Format Font
Fields
Sheet
Rows
Columns
Default
Worksheet: Tableau Boo..
Pane: Tableau Boo..
Header: Tableau Boo..
Tooltip: Tableau Book
Title: 9
Total
Pane:
Header:
Grand Total
Pane:
Header: Tableau Boo..
More colors...
Clear
Pages
Filters
Marks
Automatic
Size
Label
Tooltip
Columns
SUM(Sales)
Rows
Category
Sales
Select Color
Basic colors
Pick Screen Color
Custom colors
Add to Custom Colors
Hue: 0
Sat: 255
Val: 170
Red: 170
Green: 0
Blue: 0
HTML: #aa0000
OK
Cancel
450K 500K 550K 600K
Sales


ard Story Analysis Map Format Server Window Help
Dashboard...
Story...
Workbook...
Font...
Alignment...
Shading...
Borders...
Lines...
Animations...
Reference Lines...
Drop Lines...
Annotations...
Title and Caption...
Field Labels...
Legends...
Filters...
Standard
Pages
Filters
Marks
Automatic
Color
Size
Detail
Tooltip
M(Sales)
tegory
ory
K 100K 150K 200K 250K 300K 350K

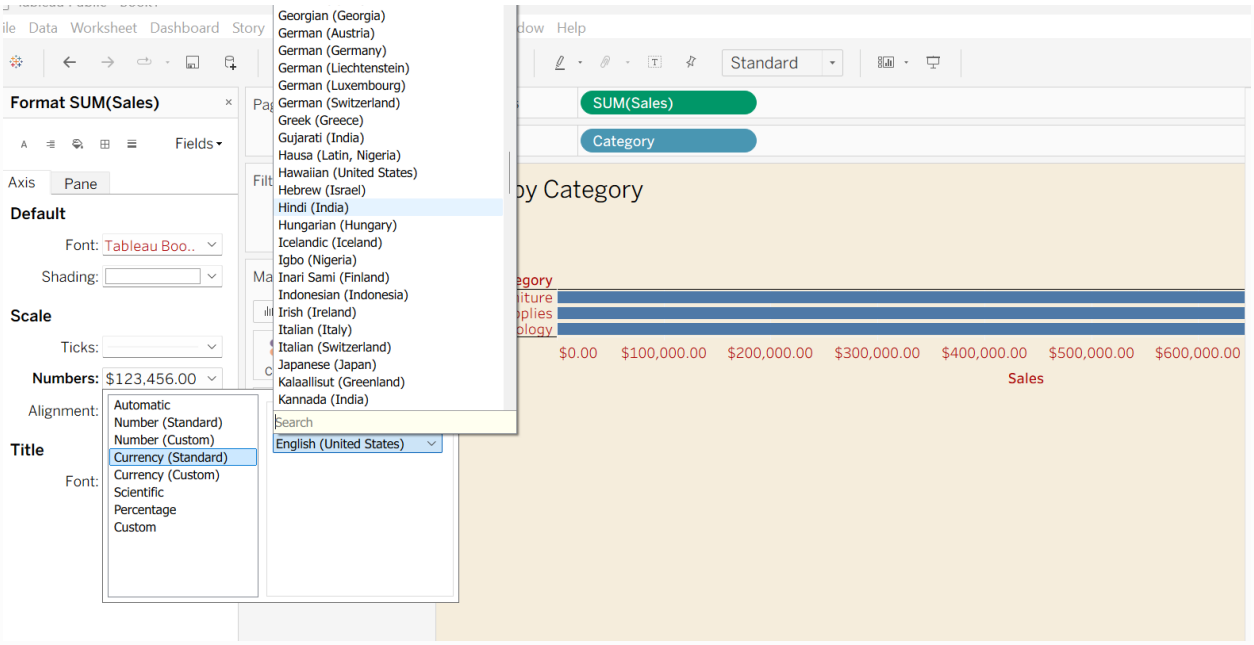

Format Shading
Fields
Sheet Rows Columns
Default
Worksheet:
Pane:
Header:
Total
Pane:
Header:
Grand Total
Pane: None
Header: None
Row Banding
Pane:
Header:
Band Size:
Level:
Clear
None
More colors...
Pages
Filters
Size
Label
Tooltip
Columns SUM(Sales)
Rows Category
Sales by Category
Category
Furniture
Office Supplies
Technology
0K 50K 100K 150K 200K 250K 300K 350K 400K 450K 500K 550K 600K
Sales

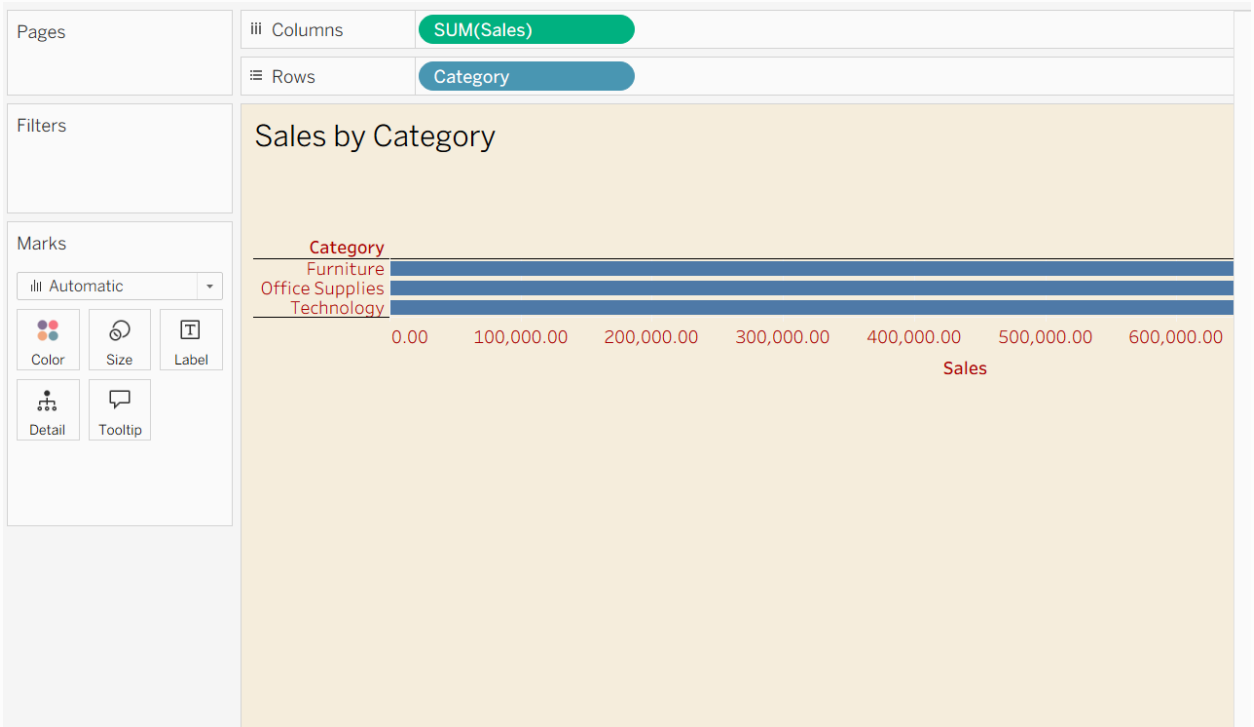

Format Borders
Fields
Sheet
Rows
Columns
Header: None
Total
Pane:
Header: None
Grand Total
Pane:
Header:
Row Divider
Pane:
Header:
Level:
Column Divi
Pane:
Header:
None
More colors...
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Columns
SUM(Sales)
Rows
Category
Select Color
Basic colors
Pick Screen Color
Custom colors
Add to Custom Colors
Hue: 0
Sat: 0
Val: 0
Red: 0
Green: 0
Blue: 0
HTML: #000000
OK
Cancel
450K
500K
550K
600K
Sales
Data Source
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Dashboard 1
Dashboard 2


Sales by Category
Category
Furniture
Office Supplies
Technology
0K
250K
300K
350K
400K
450K
500K
550K
600K
Sales
Category: Technology
Sales: 836,154
Sheet 5
Sheet 6
Dashboard 1
Dashboard 2

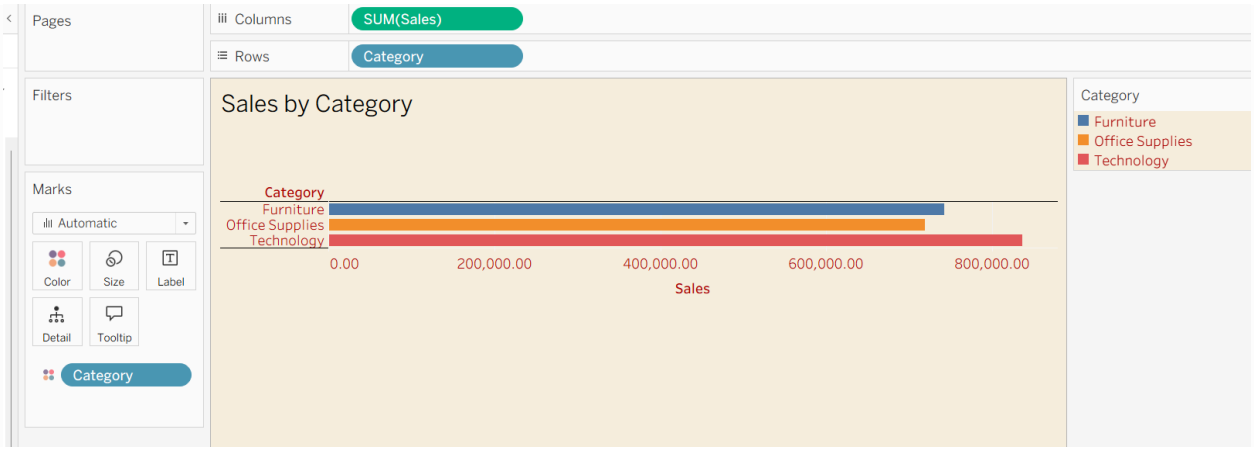


Format the Sales axis: Right-click the Sales axis and select **Format**. Set the number format to currency if appropriate.





Drag category – to Colors

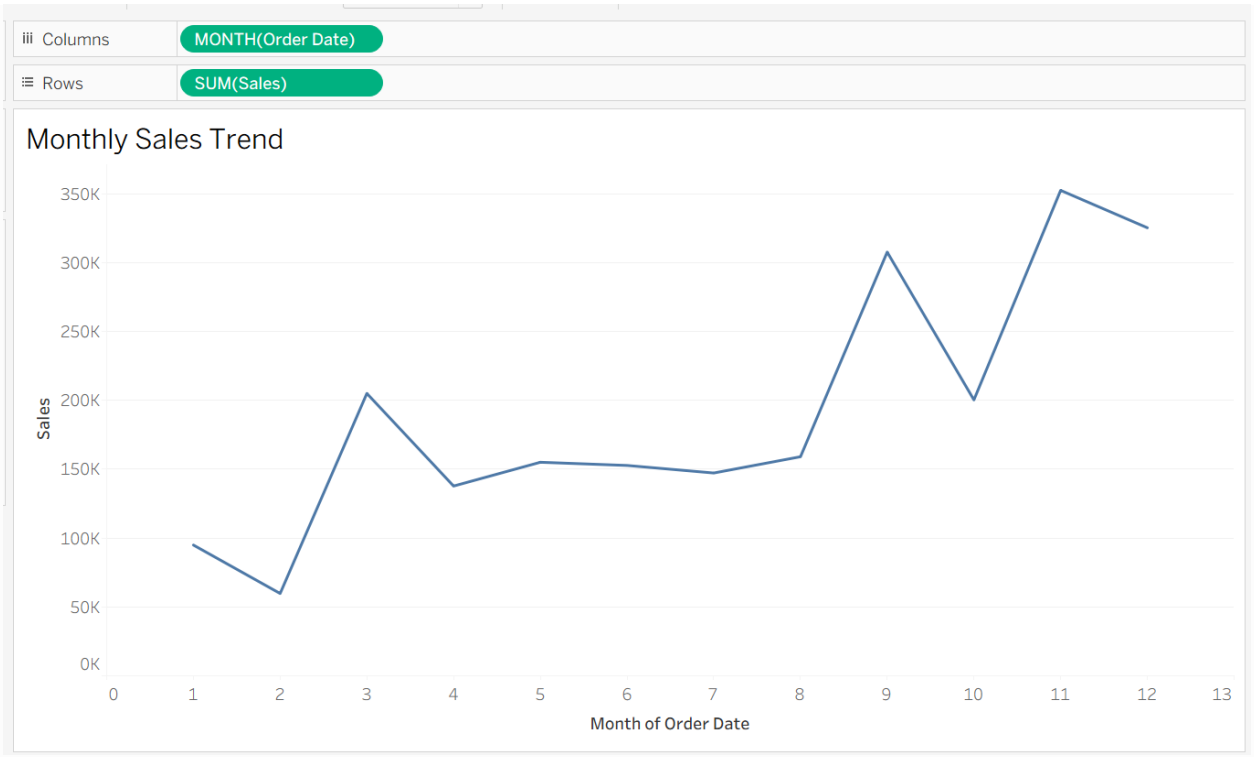


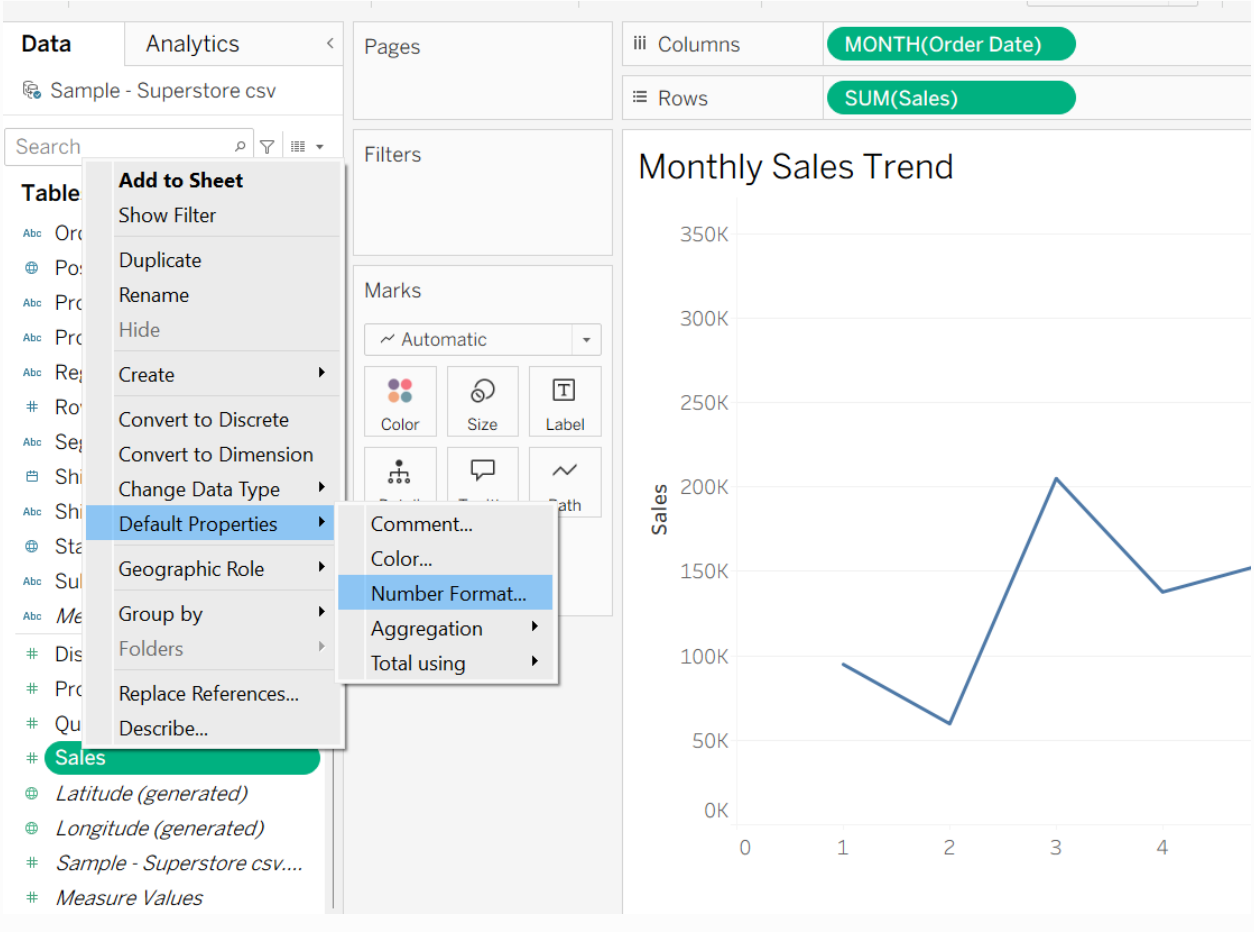
6(b) Customize Number, Date, and Null Value Formats

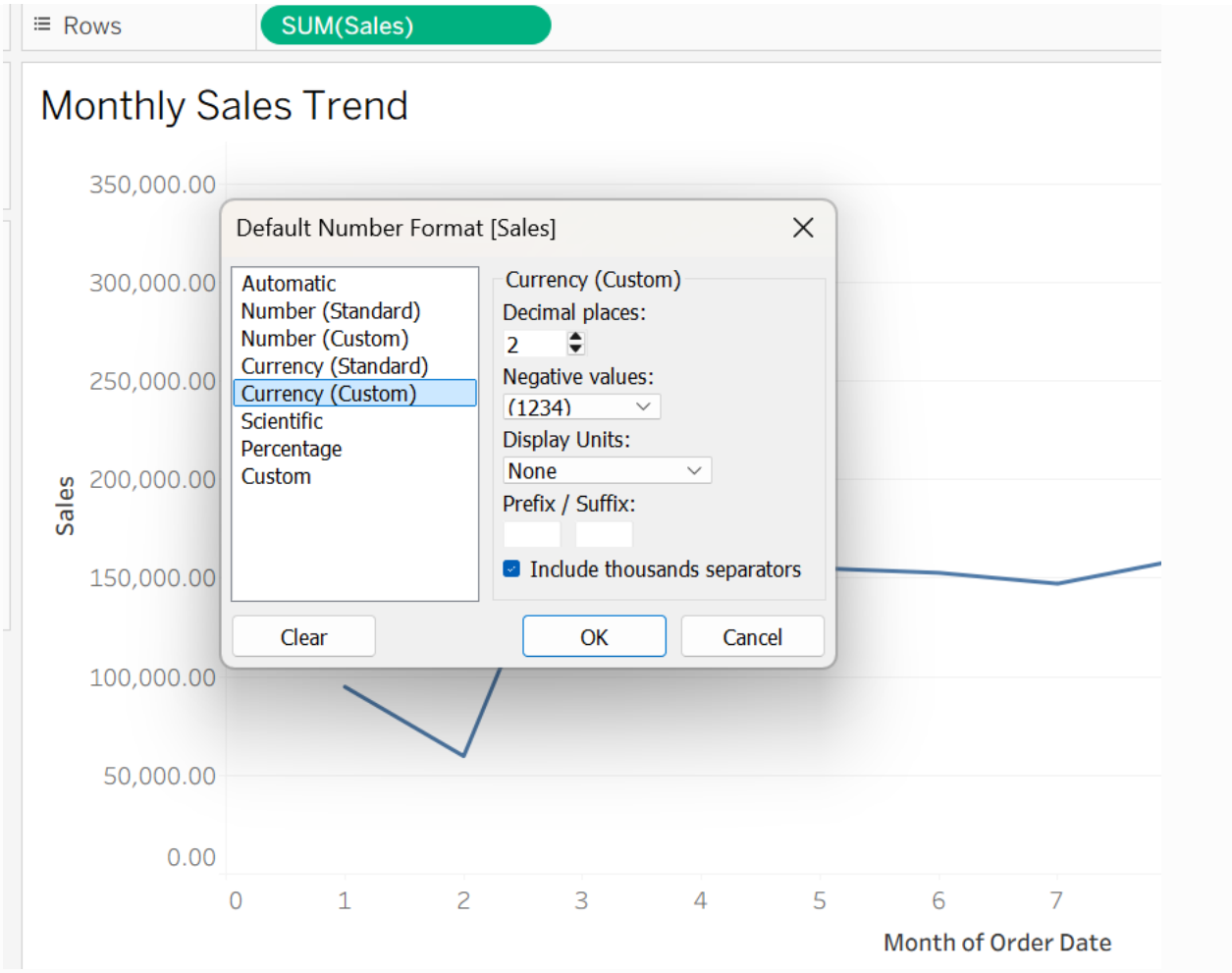
Aim: To customize number, date, and null value formats in Tableau using the Monthly Sales Trend worksheet.

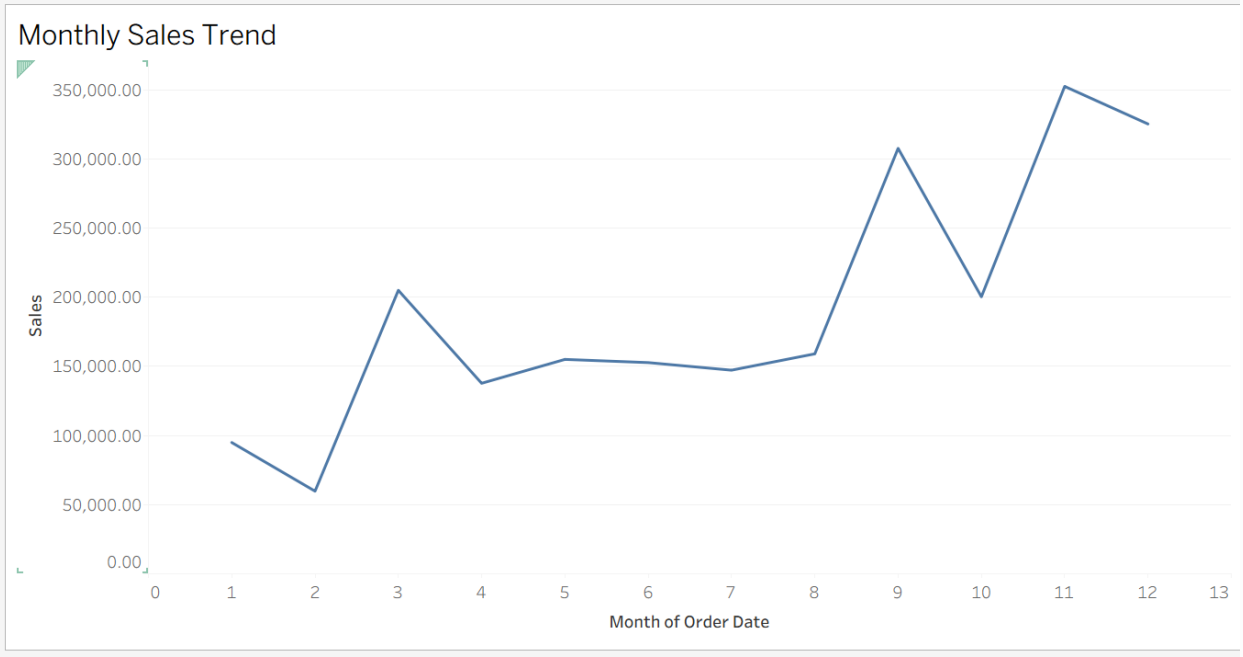
1. Number Formatting

1. Open the **Monthly Sales Trend** worksheet.
2. Right-click the **Sales** field in the Data pane.
3. Select **Default Properties → Number Format**.
4. Choose **Currency (Custom)**.
5. Set the required decimal places and currency symbol.
6. Click **OK**.


Columns
MONTH(Order Date)
Rows
SUM(Sales)
Monthly Sales Trend
350K
300K
250K
200K
150K
100K
50K
0K
Sales
0
1
2
3
4
5
6
7
8
9
10
11
12
13
Month of Order Date

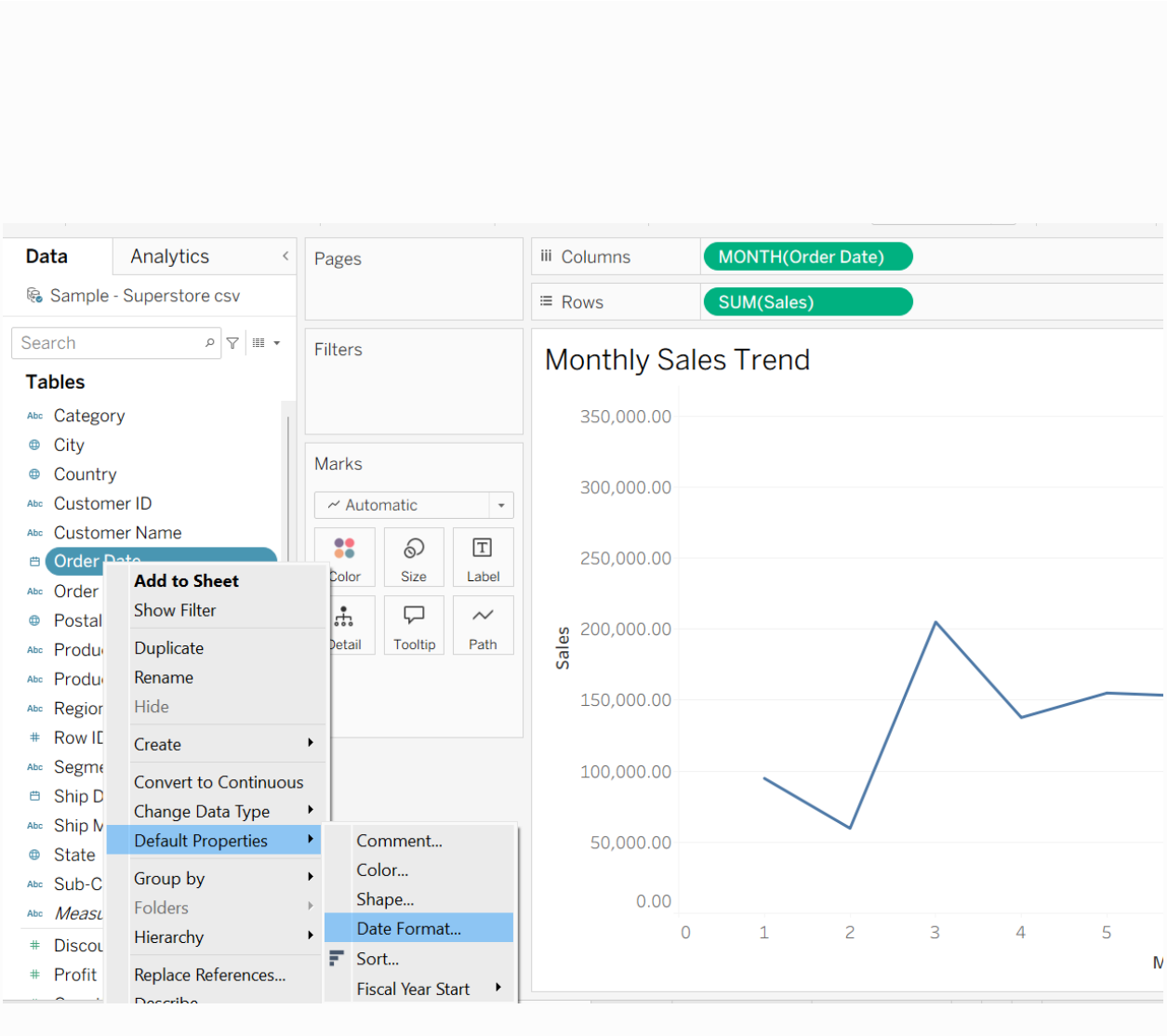

Data
Analytics
Sample - Superstore csv
Search
Table
Add to Sheet
Show Filter
Duplicate
Rename
Hide
Create
Convert to Discrete
Convert to Dimension
Change Data Type
Default Properties
Geographic Role
Group by
Folders
Replace References...
Describe...
Comment...
Color...
Number Format...
Aggregation
Total using
Sales
Latitude (generated)
Longitude (generated)
Sample - Superstore csv....
Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Columns
MONTH(Order Date)
Rows
SUM(Sales)
Monthly Sales Trend
350K
300K
250K
200K
150K
100K
50K
0K
Sales
0
1
2
3
4

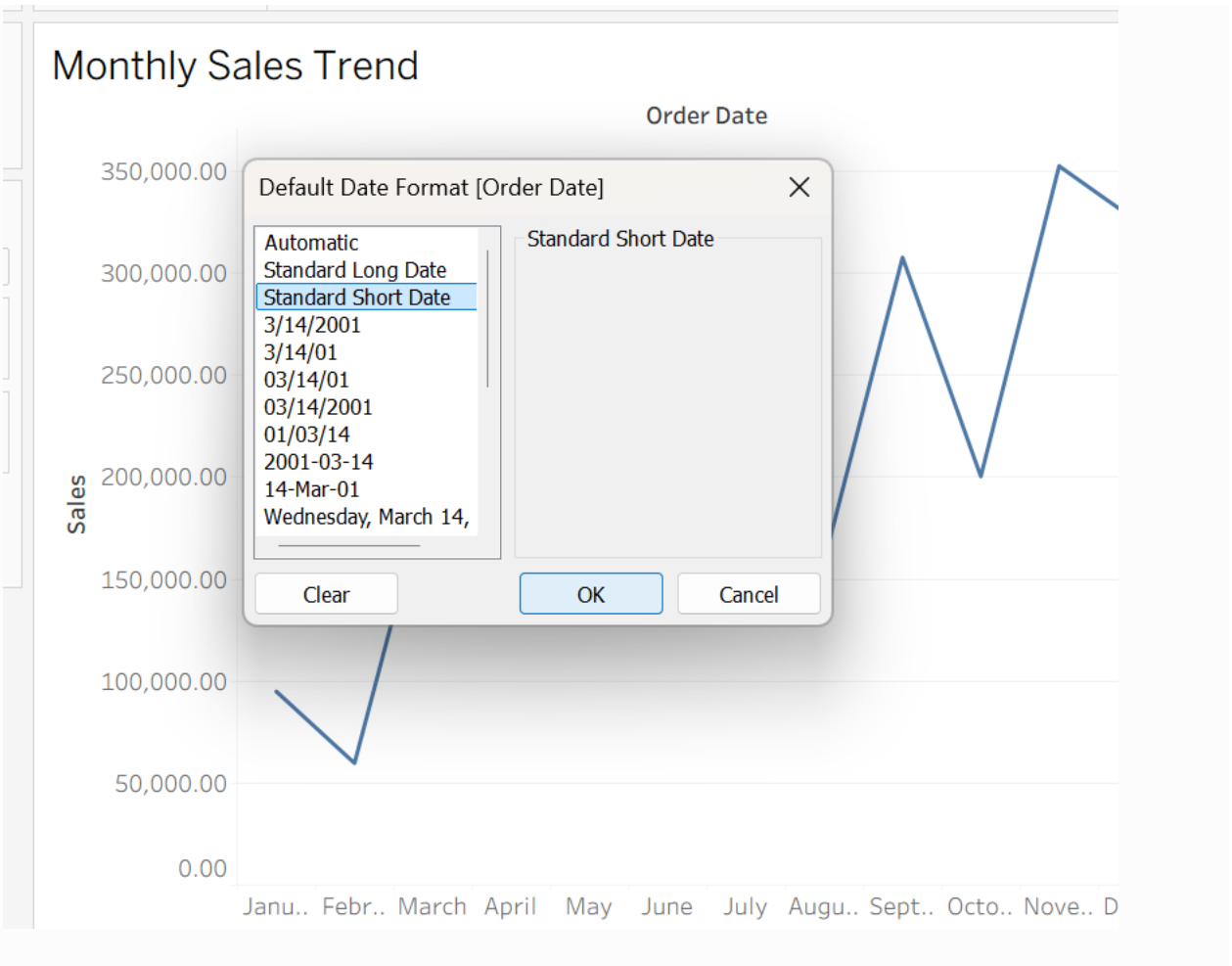

Rows
SUM(Sales)
Monthly Sales Trend
350,000.00
300,000.00
250,000.00
200,000.00
150,000.00
100,000.00
50,000.00
0.00
Sales
0
1
2
3
4
5
6
7
Month of Order Date
Default Number Format [Sales]
Automatic
Number (Standard)
Number (Custom)
Currency (Standard)
Currency (Custom)
Scientific
Percentage
Custom
Currency (Custom)
Decimal places:
2
Negative values:
(1234)
Display Units:
None
Prefix / Suffix:
Include thousands separators
Clear
OK
Cancel

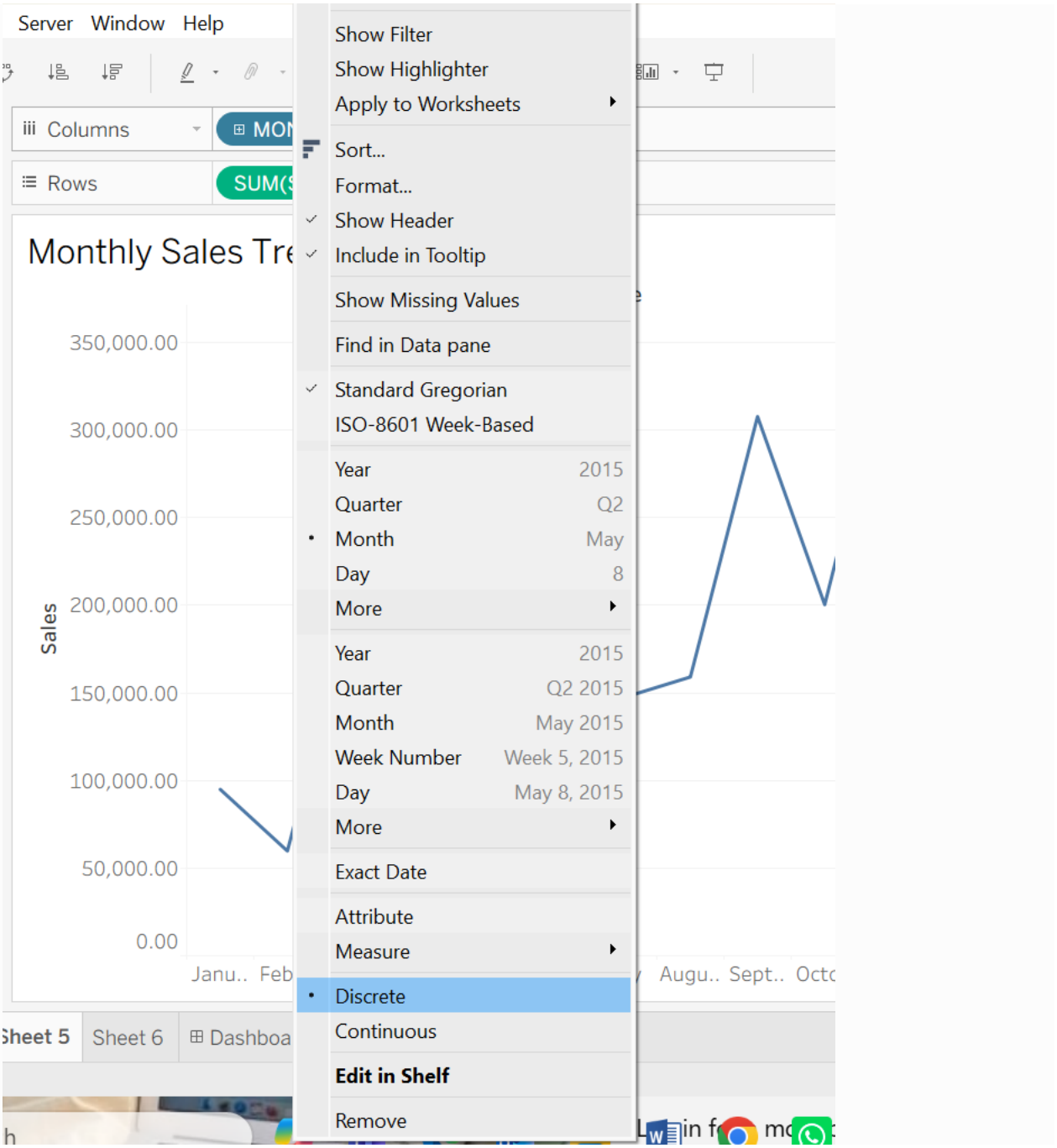

Monthly Sales Trend
350,000.00
300,000.00
250,000.00
200,000.00
150,000.00
100,000.00
50,000.00
0.00
Sales
0
1
2
3
4
5
6
7
8
9
10
11
12
13
Month of Order Date

2. Date Formatting

1. In the Monthly Sales Trend worksheet, locate the **Order Date** field in the Data pane.
2. Right-click **Order Date**.
3. Select **Default Properties → Date Format**, if available.
4. Choose the required date format, such as month and year.
5. Alternatively, right-click the date axis and select **Format** to customize the date display.
6. Check that the horizontal axis displays the months clearly.




Monthly Sales Trend
Order Date
Sales
350,000.00
300,000.00
250,000.00
200,000.00
150,000.00
100,000.00
50,000.00
0.00
Janu..
Febr..
March
April
May
June
July
Augu..
Sept..
Octo..
Nove..
D
Default Date Format [Order Date]
Automatic
Standard Long Date
Standard Short Date
3/14/2001
3/14/01
03/14/01
03/14/2001
01/03/14
2001-03-14
14-Mar-01
Wednesday, March 14,
Standard Short Date
Clear
OK
Cancel



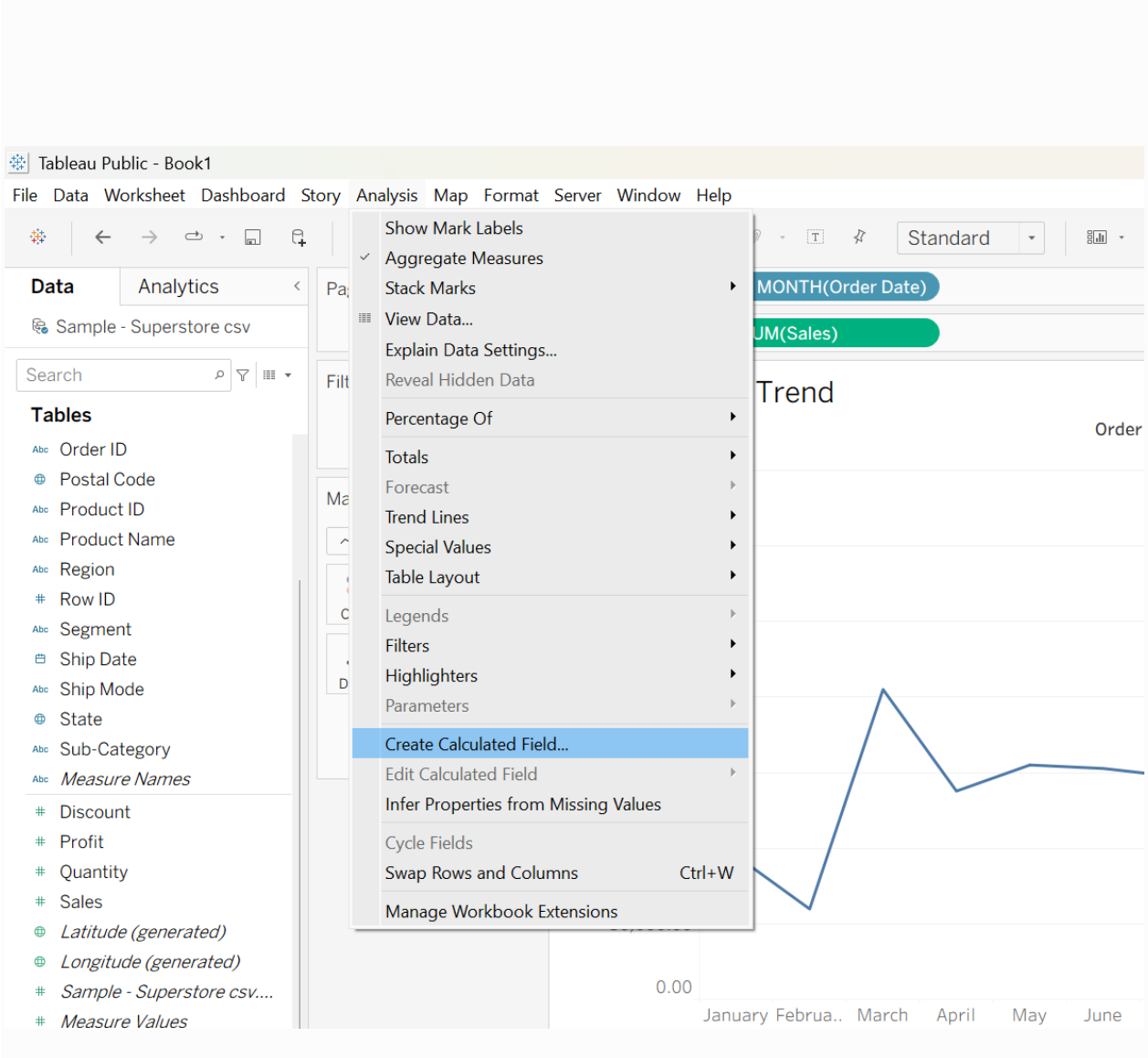
3. Null Value Formatting

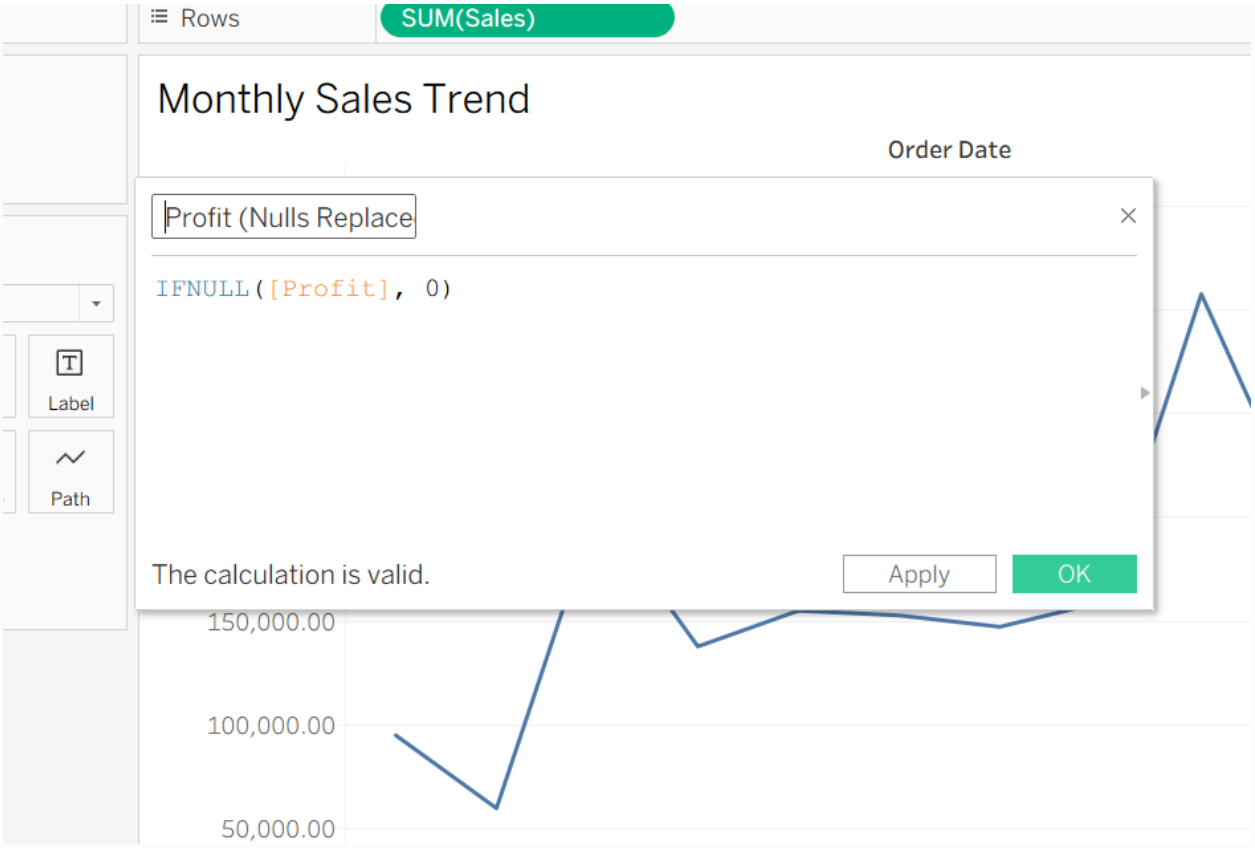
1. Open the Monthly Sales Trend worksheet and check whether any null values are displayed.
2. If a null indicator appears, click it to view the available options for handling the missing values.
3. To replace null Profit values with zero, create a calculated field by selecting **Analysis → Create Calculated Field**.
4. Name the field **Profit (Nulls Replaced)**.
5. Enter the formula:

```
IFNULL([Profit], 0)
```

6. Click **Apply**, then **OK**.
7. Use the calculated field if appropriate for the analysis.

Result: Number formats are customized, dates are displayed clearly, and null values are handled appropriately in Tableau.




Rows
SUM(Sales)
Monthly Sales Trend
Order Date
Profit (Nulls Replace
IFNULL([Profit], 0)
The calculation is valid.
Apply
OK
Label
Path
150,000.00
100,000.00
50,000.00


Columns
MONTH(Order Date)
Rows
SUM(Sales)
Monthly Sales Trend
Order Date
Sales
350,000.00
300,000.00
250,000.00
200,000.00
150,000.00
100,000.00
50,000.00
0.00
January
Februa..
March
April
May
June
July
August
Septe..
October
Novem..
Decem..
Sheet 5
Sheet 6
Dashboard 1
Dashboard 2

Practical 7: Creating Interactive Dashboards and Data Stories in Tableau

Based on your uploaded Data Visualization Practical document, the steps below follow Practical 7, sections (d) and (e), pages 51–52. The dataset used is Sample – Superstore.

Tabulae Submission-1.docx

7(a) Implement Highlight and Filter Actions

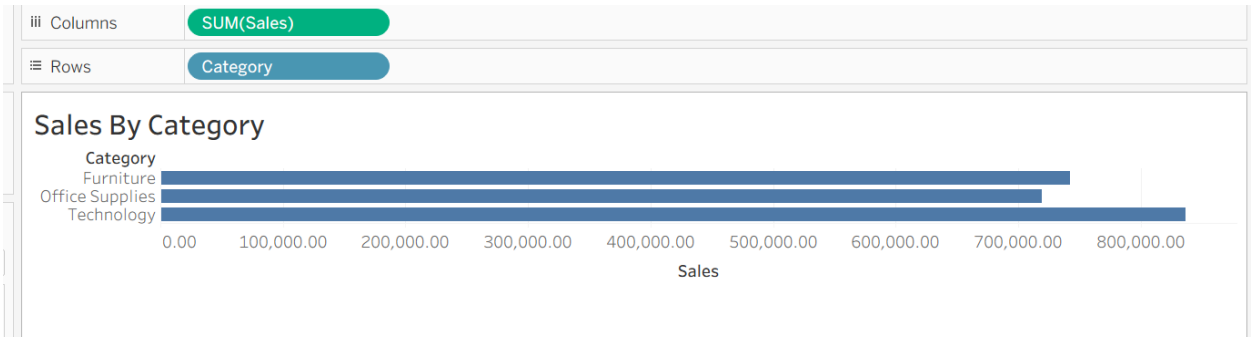
Aim: To implement Highlight and Filter Actions in Tableau for interactive data analysis

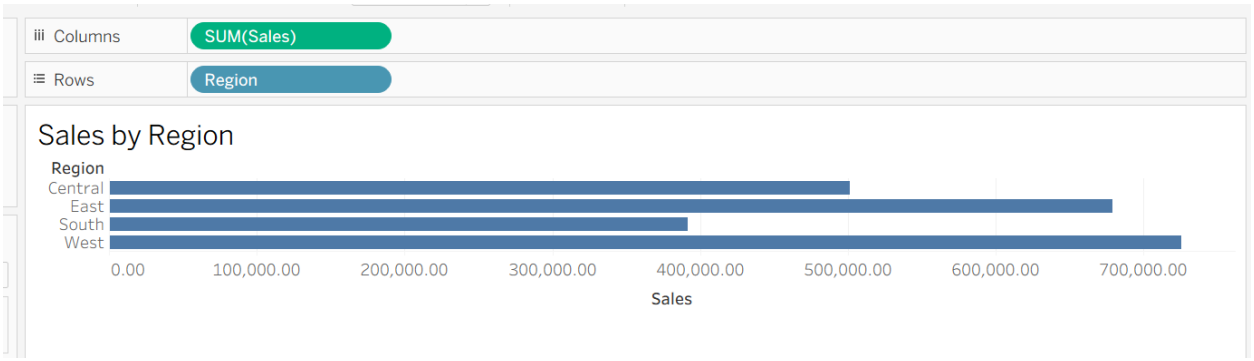
7(a) Implement Highlight and Filter Actions

Aim: To implement Highlight and Filter Actions in Tableau for interactive data analysis.

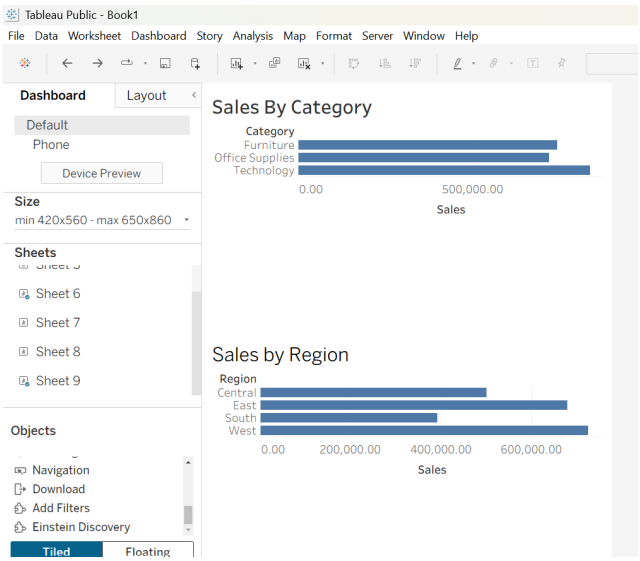
Procedure

1. Open Tableau and connect to the Sample – Superstore dataset.
2. Create worksheets such as Sales by Category and Sales by Region.

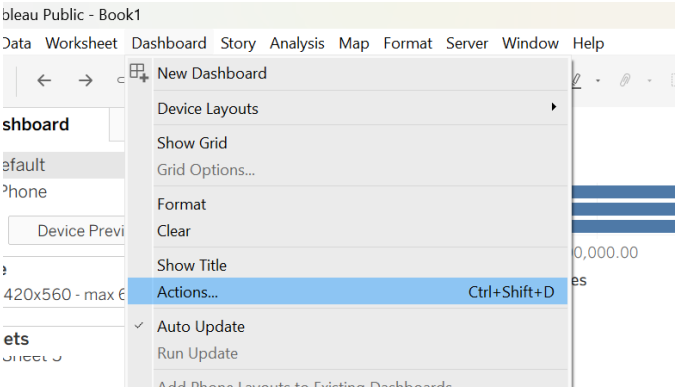




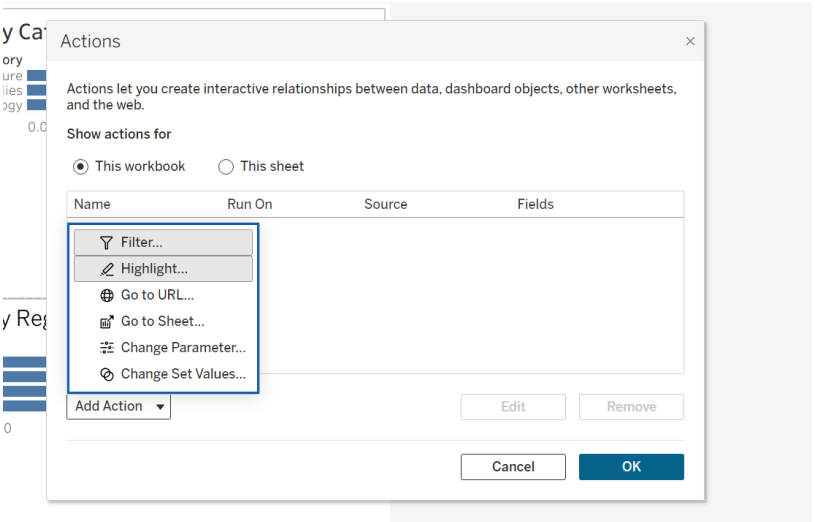
3. Click New Dashboard and drag the worksheets onto the dashboard.



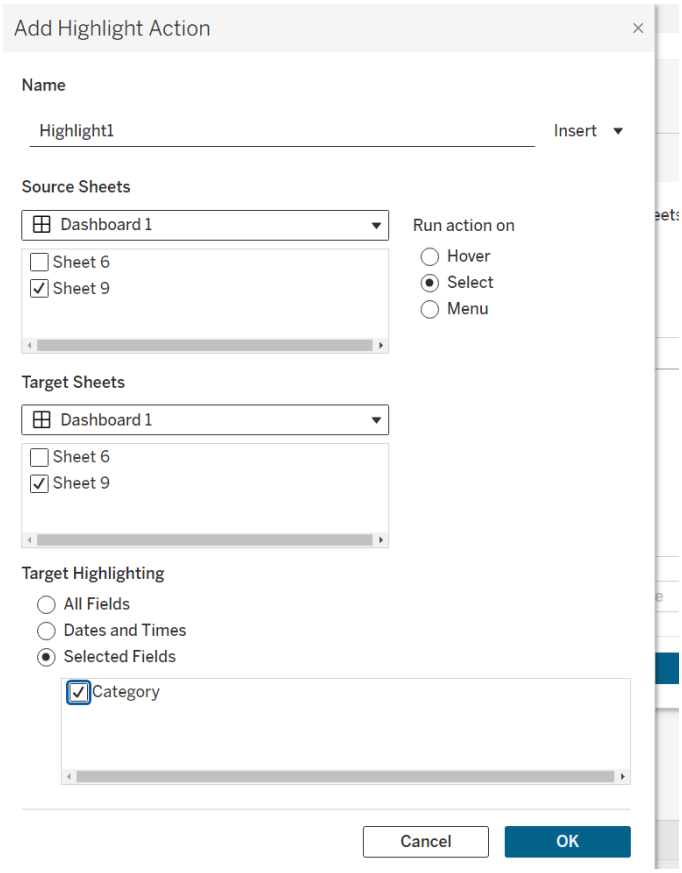
4. From the top menu, select Dashboard → Actions.



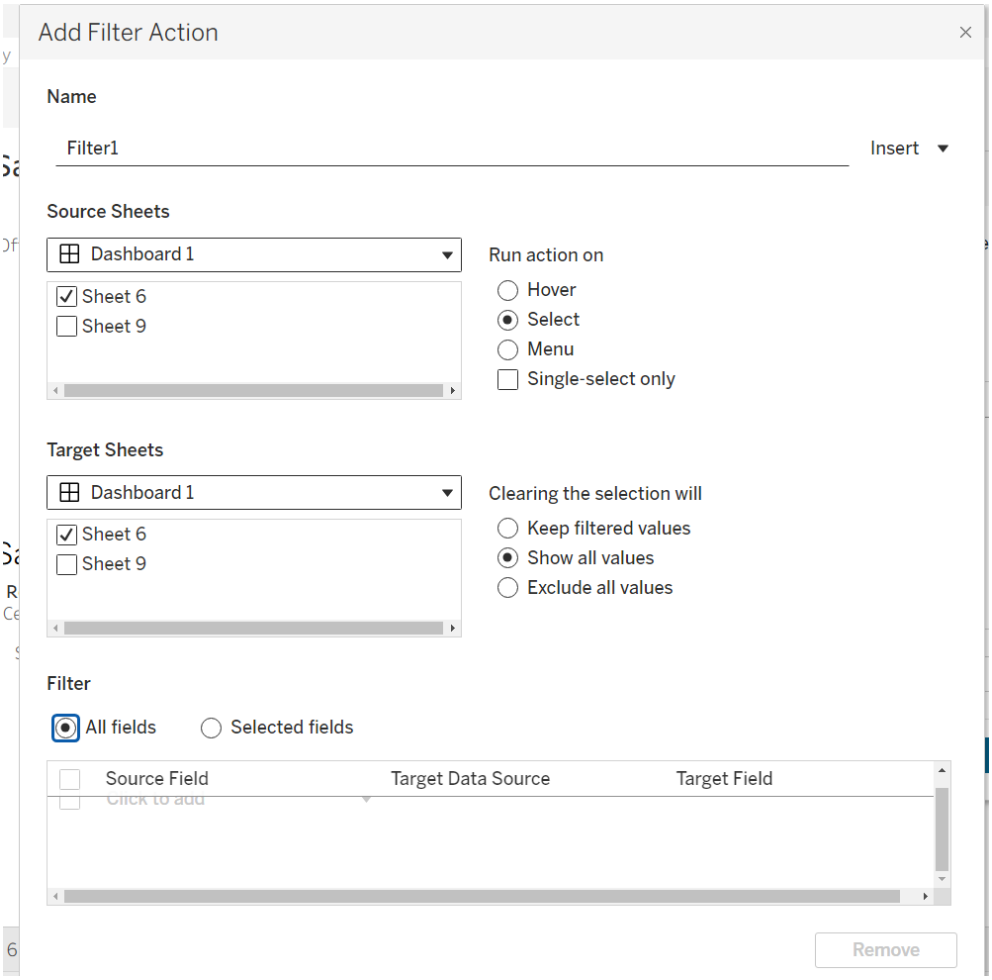
5. Click Add Action → Highlight.



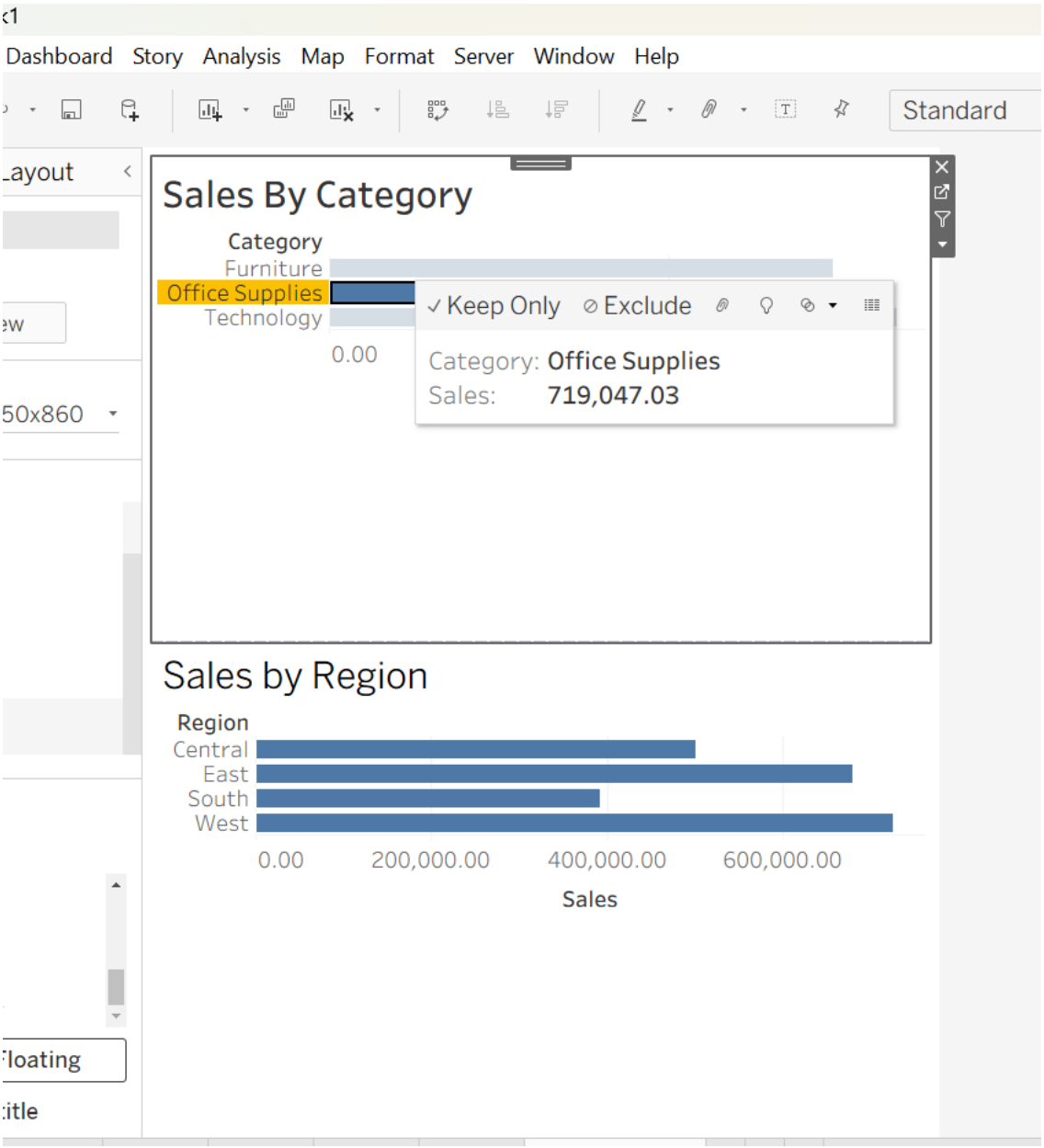
6. Select the source sheet as Sales by Category.
7. Select the target sheets where the highlighting should appear.
8. Under Target Highlighting, select Selected Fields and choose `Category`.
9. Click OK to save the Highlight Action.

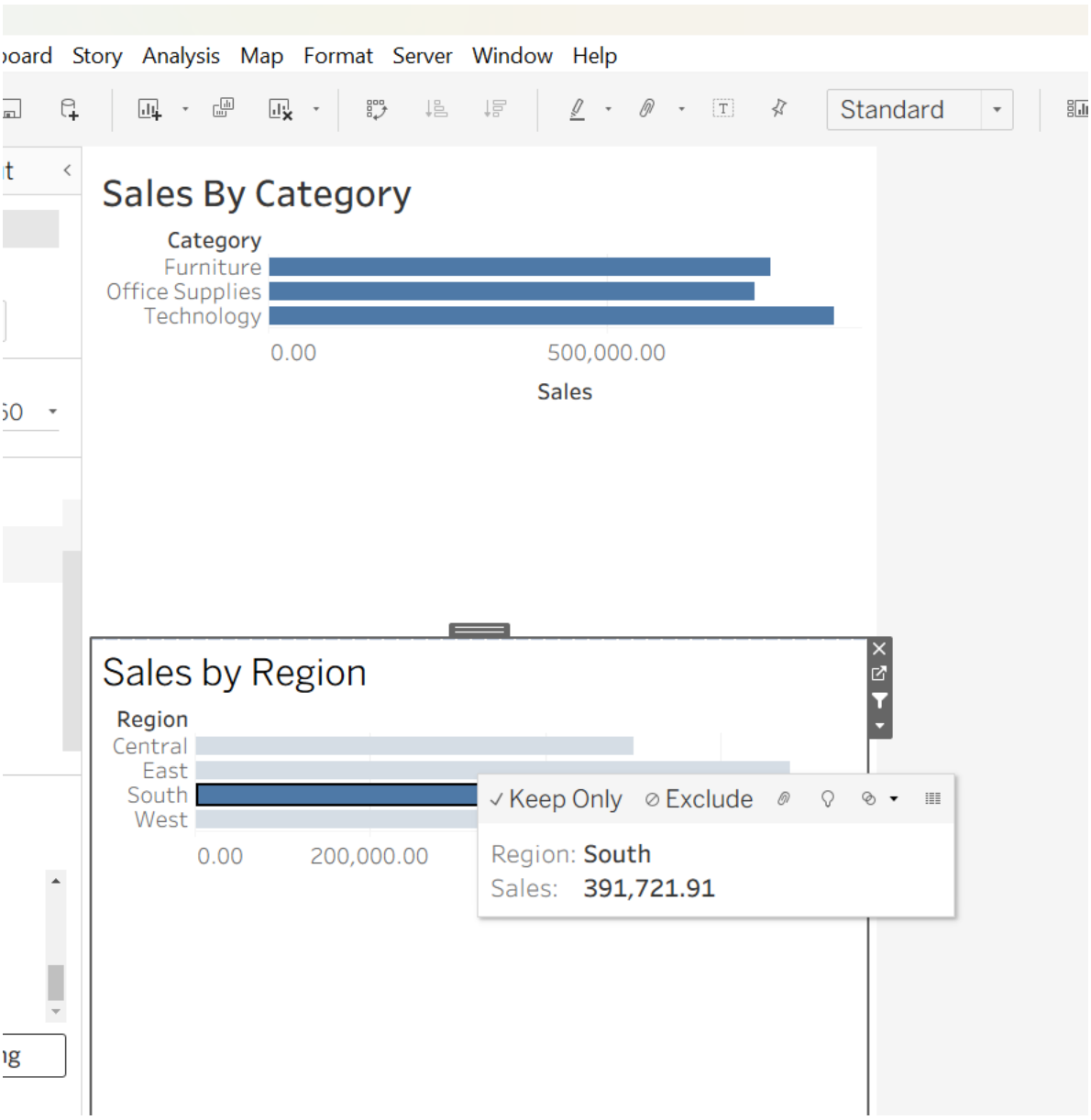


10. Again, go to Dashboard → Actions → Add Action → Filter.
11. Select Sales by Region as the source sheet and choose the target sheets.
12. Set Run action on to Select.
13. Click OK and close the Actions dialog box.



14. Click a category or region in the dashboard to observe the highlighting or filtering.


Dashboard Story Analysis Map Format Server Window Help
Standard
Sales By Category
Category
Furniture
Office Supplies
Technology
0.00
Keep Only
Exclude
Category: Office Supplies
Sales: 719,047.03
Sales by Region
Region
Central
East
South
West
0.00
200,000.00
400,000.00
600,000.00
Sales



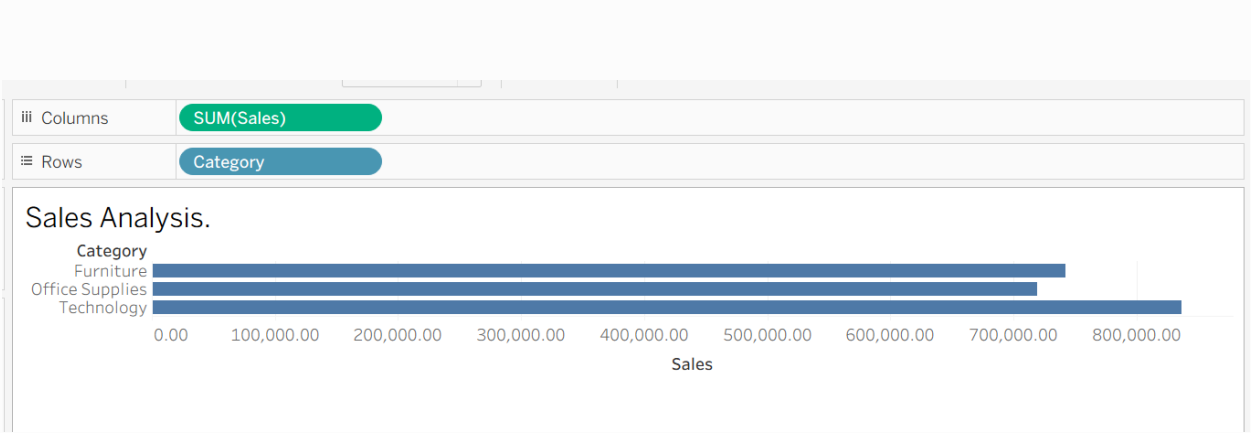
Output: Selecting a category highlights related data, while selecting a region filters the target visualizations.

7(b) Create Dynamic Dashboards Using Parameters

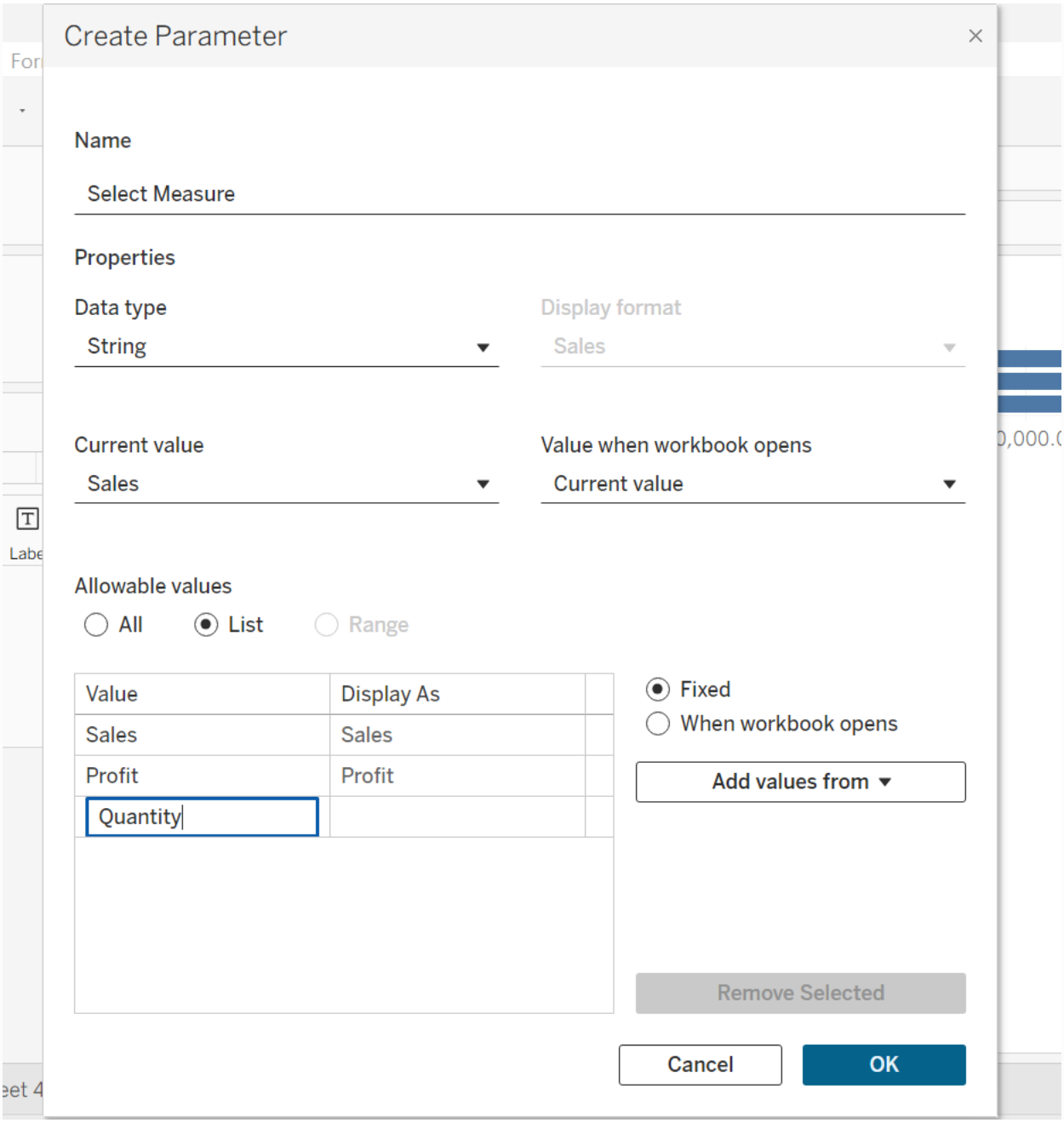
Aim: To create a dynamic Tableau dashboard using parameters that allow users to change the displayed analysis interactively.

Procedure

1. Open Tableau and connect to the Sample – Superstore dataset.
2. Create a new worksheet named Sales Analysis.
3. Drag `Category` to Rows and `Sales` to Columns to create a bar chart.



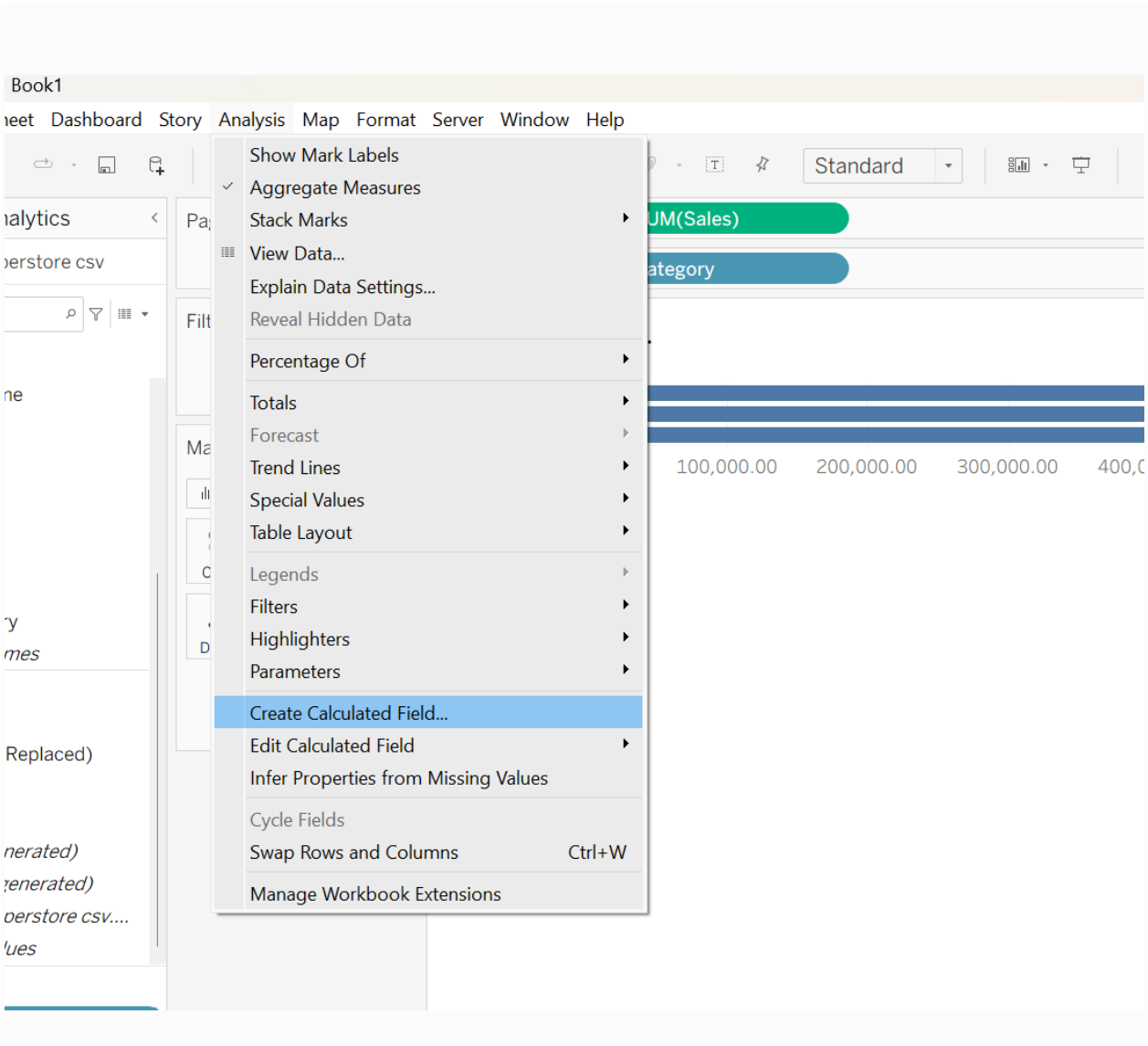
- In the Data pane, right-click and select Create → Parameter.
- Configure the parameter as follows:
  - Name: `Select Measure`
  - Data Type: String
  - Allowable Values: List
  - Values: `Sales`, `Profit`, `Quantity`
- Click OK.

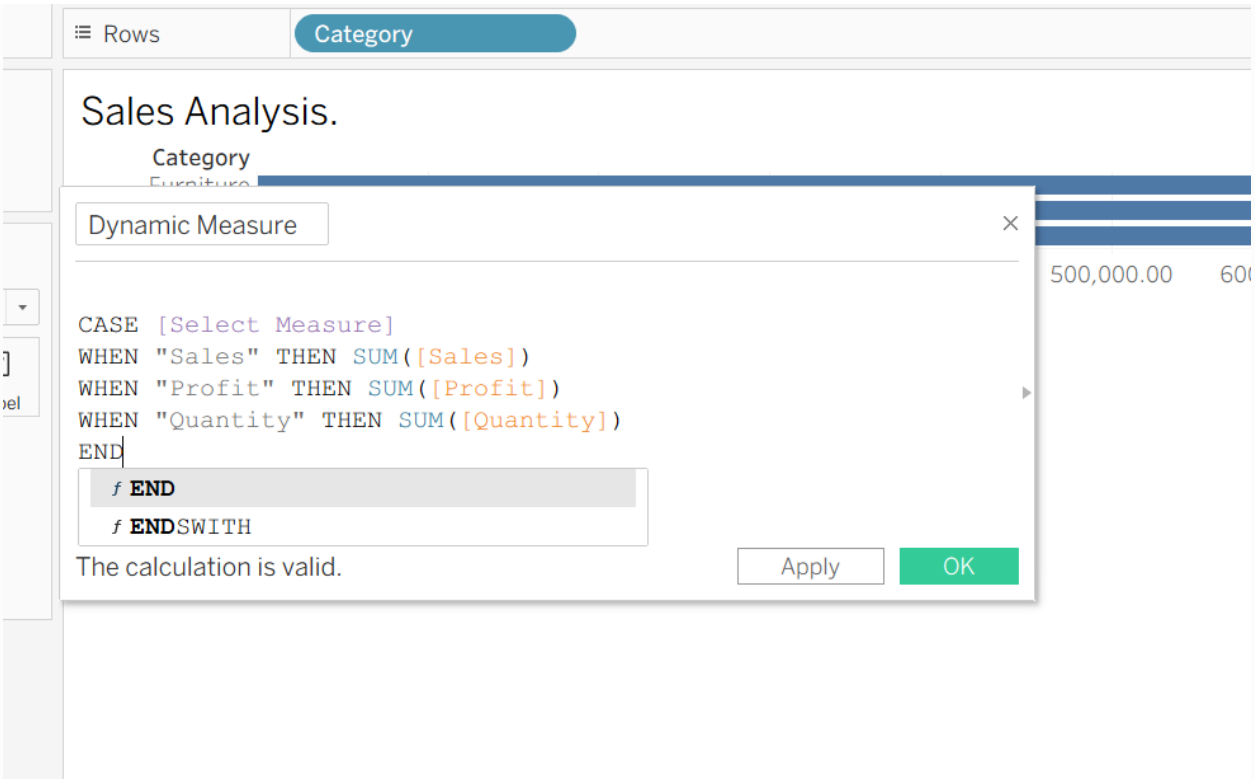


- Right-click the Data pane and select Create → Calculated Field.
- Name the calculated field `Dynamic Measure` and enter this formula:

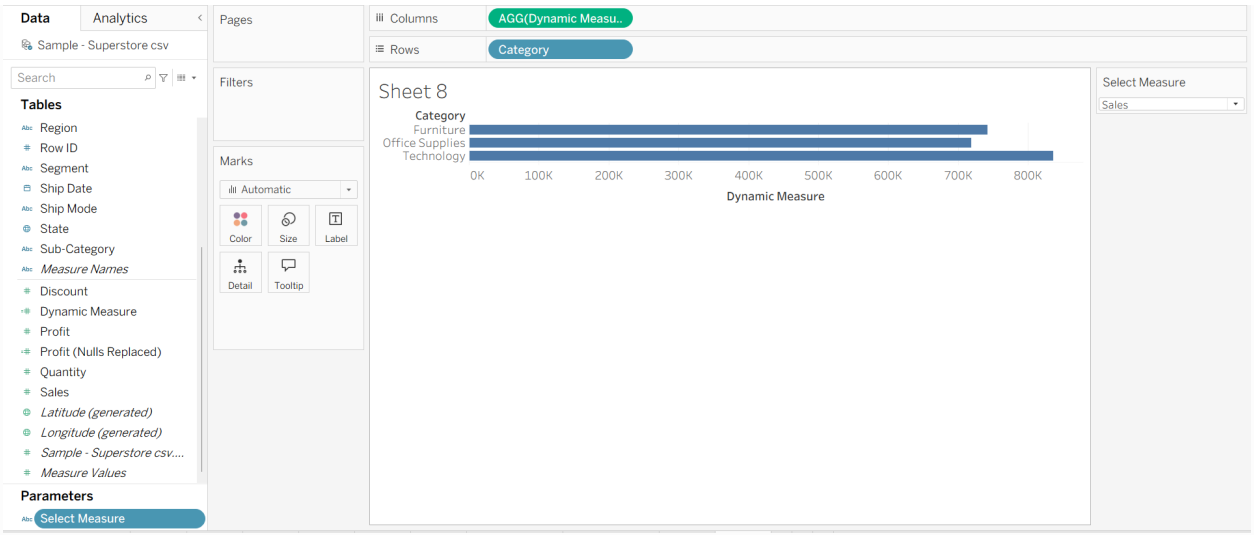
```
CASE [Select Measure]
WHEN "Sales" THEN SUM([Sales])
WHEN "Profit" THEN SUM([Profit])
WHEN "Quantity" THEN SUM([Quantity])
END
```

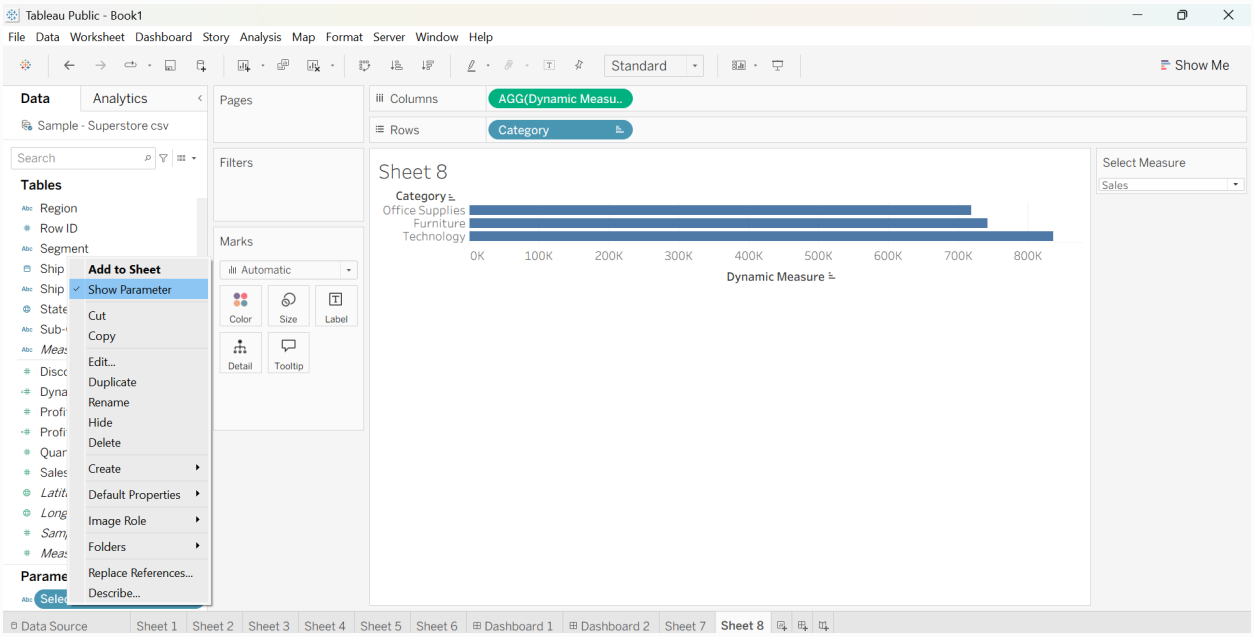
- Click OK.

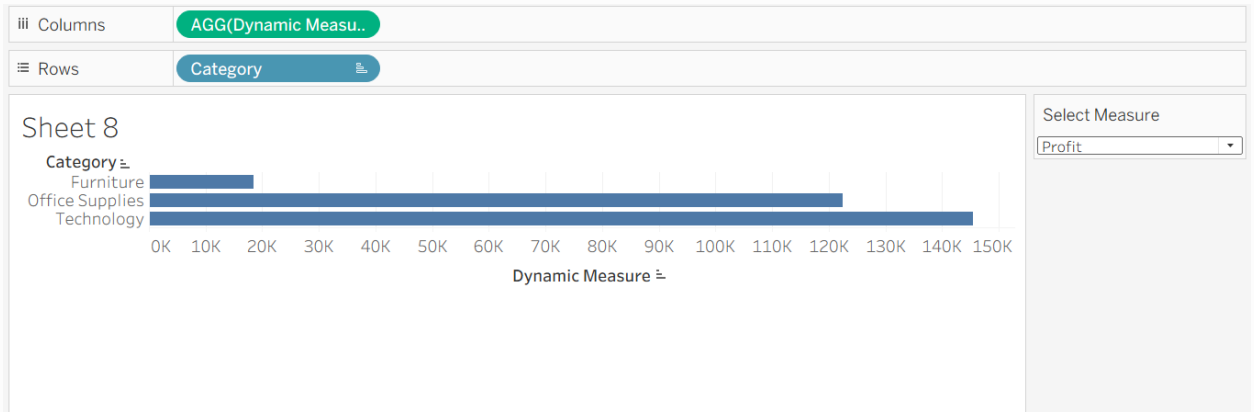

Book1
Dashboard
Story
Analysis
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Format
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Window
Help
Standard
Show Mark Labels
Aggregate Measures
Stack Marks
View Data...
Explain Data Settings...
Reveal Hidden Data
Percentage Of
Totals
Forecast
Trend Lines
Special Values
Table Layout
Legends
Filters
Highlighters
Parameters
Create Calculated Field...
Edit Calculated Field
Infer Properties from Missing Values
Cycle Fields
Swap Rows and Columns Ctrl+W
Manage Workbook Extensions
100,000.00
200,000.00
300,000.00

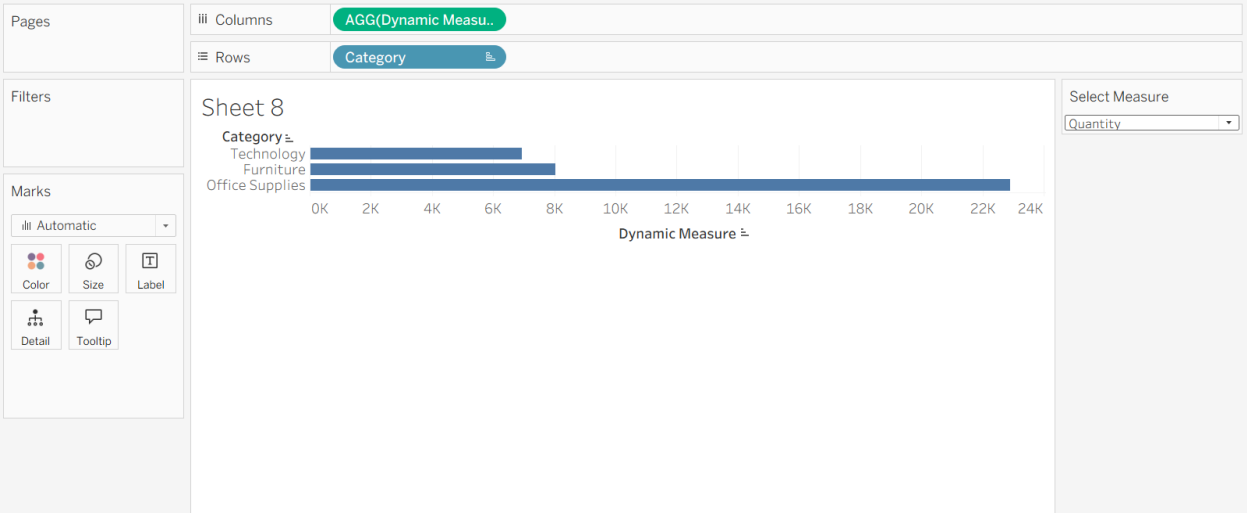


- Open a new worksheet.
- Drag Category to Rows.
- Drag Dynamic Measure to Columns.
- You should see a bar chart showing the selected measure.
- If the parameter control is not visible, right-click `Select Measure` in the Data pane and select Show Parameter.


Data
Analytics
Sample - Superstore csv
Search
Tables
Region
Row ID
Segment
Ship Date
Ship Mode
State
Sub-Category
Measure Names
Discount
Dynamic Measure
Profit
Profit (Nulls Replaced)
Quantity
Sales
Latitude (generated)
Longitude (generated)
Sample - Superstore csv....
Measure Values
Parameters
Select Measure
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Columns
AGG(Dynamic Measu..
Rows
Category
Sheet 8
Category
Furniture
Office Supplies
Technology
0K
100K
200K
300K
400K
500K
600K
700K
800K
Dynamic Measure
Select Measure
Sales


Tableau Public - Book1
File
Data
Worksheet
Dashboard
Story
Analysis
Map
Format
Server
Window
Help
Standard
Show Me
Data
Analytics
Sample - Superstore csv
Search
Tables
Region
Row ID
Segment
Add to Sheet
Show Parameter
Cut
Copy
Edit...
Duplicate
Rename
Hide
Delete
Create
Default Properties
Image Role
Folders
Replace References...
Describe...
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Columns
AGG(Dynamic Measu..
Rows
Category
Sheet 8
Category
Office Supplies
Furniture
Technology
0K
100K
200K
300K
400K
500K
600K
700K
800K
Dynamic Measure
Select Measure
Sales
Data Source
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Dashboard 1
Dashboard 2
Sheet 7
Sheet 8


Columns
AGG(Dynamic Measu..
Rows
Category
Sheet 8
Category
Furniture
Office Supplies
Technology
0K
10K
20K
30K
40K
50K
60K
70K
80K
90K
100K
110K
120K
130K
140K
150K
Dynamic Measure
Select Measure
Profit



Step 3: Test the parameter

Change the `Select Measure` control between:

- Sales — displays sales by category.
- Profit — displays profit by category.
- Quantity — displays quantity by category.

The chart should update automatically when you change the selection

7(c) Use URL Actions for Interactivity

Aim: To implement URL Actions in Tableau to open an external webpage based on the selected data.

Procedure

1. Open Tableau and connect to the Sample – Superstore dataset.
2. Create a worksheet named Product Analysis.
3. Drag `Product Name` to Rows and `Sales` to Columns to create a bar chart.
4. Click New Dashboard and drag the Product Analysis worksheet onto it.
5. From the top menu, select Dashboard → Actions.
6. Click Add Action → Go to URL.
7. Enter the action name as `Search Product`.
8. Select Product Analysis as the source sheet.
9. Under Run action on, select Select.
10. In the URL field, enter:

```
https://www.google.com/search?q=<Product Name>
```

11. If prompted, select the appropriate field from the Insert menu so that the selected product name is inserted into the URL.
12. Click OK, then click OK again to close the Actions dialog.
13. Run the dashboard and click a product bar.
14. Tableau opens the corresponding Google search results in a browser.

Output: Clicking a product in the dashboard opens a webpage displaying search results for that product.

Result

Successfully implemented Highlight and Filter Actions, created a dynamic dashboard using parameters, and used URL Actions to make the Tableau dashboard interactive.